A MACHINE-VERIFIED RESEARCH MONOGRAPH

SEPTINAL

Formal Choptuik & the Septinial Braking Satellite — two Lean 4 libraries, one rational certificate bridge, and the constants of critical collapse read off the Klein quartic



Around the threshold of gravitational collapse one number — the growth rate λ_0 of a single unstable mode — fixes the exponent γ of the black-hole mass scaling law. This monograph documents a research package that turns that bookkeeping into kernel-checked theorems and then reads the same constants from the {7,3} geometry of the Klein quartic curve: a 24-slot dial with one structural gap, $24 - 2 = 22 = b_2(K3)$.

0 SORRY	0 PROJECT AXIOMS	145 THEOREMS	95 AUDITED · EXIT 0	22 = b_2(K3)

Ishak Khamzatovich Isaev

ORCID 0009-0003-7299-0701 · NALCHIK, KABARDINO-BALKARIAN REPUBLIC

WING OF CHOPTUIK_AC_BC · SEED 1993

MMXXVI

§0 Preflight — everything in three commands

This monograph is the printable face of a machine-checked artifact. Before reading a single theorem, the skeptical reader can replay the whole stack: the Lean build (2 894 proof steps, exit 0), the rational certificates (emitted and re-checked on every run), and the seven-task numerical harness in both language twins. Three commands from the package root are the whole ceremony; every number, figure and table in this book regenerates from them, and from no other source.

```
cd lean && lake build                                  # the proofs: 2 894 steps, exit 0
python3 scripts/certify.py emit && python3 scripts/certify.py check
python3 scripts/septinal_verify.py --ci --preset quick --task all   # 7/7 PASS
```



Fig. 0.1 — The four doors of reproduction: the Lean build, the certificate cycle, the numerical harness, and the pre-push gate; all four open onto the same gold line of committed artifacts. Panels 2, 3, 4, 7 — the (2,3,7) signature.

The rest of this preface explains what each layer is allowed to claim. The rule of the house is printed on the cover: *machine-readable or it did not happen* — a claim that cannot be re-derived from the artifact is marked INTERP and is never allowed to masquerade as a theorem.

FRONT MATTER

Machine-Readable or It Did Not Happen

That single sentence is the constitution of everything documented in this monograph. The Septinal package does not report results that a reader must take on faith: every headline claim is either a Lean 4 proof whose dependency tree the kernel prints on demand, a rational certificate re-checked fraction-by-fraction, or a number recomputed byte-identically from a fixed seed. Where a statement cannot be proved, the package does not stay silent — it states the boundary as a theorem of its own. The chapters that follow walk through the physics, the formalization, the bridge that carries numerics into the kernel, the seven large-matrix verification tasks, and the septinial geometry that closes the arc.

Abstract

Septinal is a machine-verified research package built from two Lean 4 + Mathlib libraries. The first, `Choptuik`, formalizes the bookkeeping of critical gravitational collapse for the Einstein–scalar system: the discretely self-similar (DSS) critical solution, its determination by data on a fundamental strip, uniqueness up to scale and time translation, a quantitative Newton–Kantorovich existence argument with contraction constants $\beta = 50/39$ and $L = 3/50$, a certified spectral picture with exactly one unstable mode $\lambda_0 = 2.6745$, and the mass scaling law $\gamma = 1/\lambda_0 = 2000/5349$ with the proved inequality $|\gamma - 0.374| = \frac{263}{2674500} < 10^{-3}$. The second, `Satellite`, formalizes a geometric reading of the same constants on the Klein quartic curve {7,3}: a 24-slot septinial dial whose antipodal gap {0, 12} yields the counting identity $k = 24 - 2 = 22 = b_2(\mathrm{K3})$, the chiral split 22 = 11 + 11, the symmetry cascade $48 \to 4 \to 2$, and the exact parameter-free record $\gamma_3 = 1 - \cos(2\pi/7) - \pi/1200 = 0.373892\ldots$, within $1.1 \cdot 10^{-4}$ of the numerically calibrated exponent. The two libraries are joined by a rational certificate bridge: a Python producer emits exact rational enclosures, `norm_num` re-checks every fraction inside Lean, and discharge theorems lift them to physics-level headlines. The audit covers 95 headline declarations across 24 modules (5108 lines of Lean) with zero `sorry` and zero project axioms; the only axioms in any dependency tree are Lean's three standards.

How to read this book

Every substantive claim in the chapters below carries a level badge. The badge tells you where the statement lives in the verification stack, and therefore how much machinery stands behind it:

Table 0.1 — Claim levels used throughout the monograph

BADGE	MEANING	ENFORCED BY
KERNEL	Proved inside Lean 4; the kernel checks the proof object on every build.	`lake build`, `#print axioms` audit
CERT	Exact rational certificate produced by Python, re-verified fraction-by-fraction inside Lean.	`Bridge/Generated.lean`, `certify.py check`
NUM	Numerical verification on large synthetic matrices by the stdlib-only harness twins.	`septinal_verify.py` / `.jl`
INTERP	Interpretation: a physical or geometric metaphor whose strict mathematical content is spelled out separately.	explicit lemma pointers

The command-line harness that produces all NUM-level evidence is deliberately self-contained — one file per language, standard library only. Its three modes are worth memorizing before Chapter IV dissects each task: an interactive menu (`python3 septinal_verify.py`), a silent library load (`import septinal_verify`), and a CI batch mode (`--ci --preset quick --task all`), with Julia twin `septinal_verify.jl` accepting the same grammar:

```
$ python3 septinal_verify.py --task all --preset quick     # 7 tasks, two-pass
$ python3 septinal_verify.py --ladder all                  # convergence ladder
$ python3 septinal_verify.py --task 7 --reorth none        # O(1)-memory Lanczos
$ python3 septinal_verify.py --self-test --props --contract
```

The constants of the package

Eight numbers anchor the whole narrative. Each is pinned by an exact rational record or a closed form, each carries a proved or bracketed distance to its published counterpart, and each is regenerated from seed 1993 by `scripts/compute_all.py` into `data/constants.json`. They are collected here once and referenced throughout.

Table 0.2 — The eight constants (from data/constants.json)

CONSTANT	EXACT VALUE	DECIMAL	STATUS
Unstable mode λ_0	certified [2.6744999, 2.674500101]	2.6745	KERNEL + CERT
Scaling exponent $\gamma = 1/\lambda_0$	2000/5349	0.373901663862	$\lvert\gamma - 0.374\rvert$ = 263/2674500
Echoing period Δ	1722726201/500000000	3.445452402	$\lvert\Delta - 3.4453\rvert$ = 76201/$5\cdot10^8$
Echo spacing $2\pi/\ln 2$	closed form	9.064720283654	KERNEL (echo family)
Wiggle period $\lambda_0\Delta/2$	product	4.607431224575	$\in$ [4.60, 4.62] proved
Septinial record γ_3	1 - cos(2π/7) - π/1200	0.373892204263	KERNEL, parameter-free
Sector-C constant b_Ch	1 - cos(2π/7)	0.376510198141	bracket (1−√2/2, 1/2)
Dial capacity k	24 - 2	22 = b_2(K3)	KERNEL counting theorem

Notation

The following symbols are used with fixed meanings everywhere below. $D = \mathbb{Z}/24\mathbb{Z}$ is the dial group of 24 slots; the antipodal gap is $G = \{0, 12\}$ with shift $\sigma(x) = x + 12$; the two chiral arcs are $A = \{1, \ldots, 11\}$ and $B = \{13, \ldots, 23\}$; $V_4 = \{\mathrm{id}, +12, -x, 12-x\}$ is the Klein four-group stabilizing the gap inside the dihedral group D_{24} of order 48. On the physics side, p is the scalar-field amplitude, p^* the critical amplitude, M_{BH} the black-hole mass, λ_0 the single unstable growth rate, Δ the echoing period, and $\gamma = 1/\lambda_0$ the scaling exponent. Lean identifiers are set in monospace exactly as they appear in the sources — for instance

`Satellite.Dial.card_dial` or `Scaling.gamma_close_to_published` — so that every claim can be located by search.

Contents

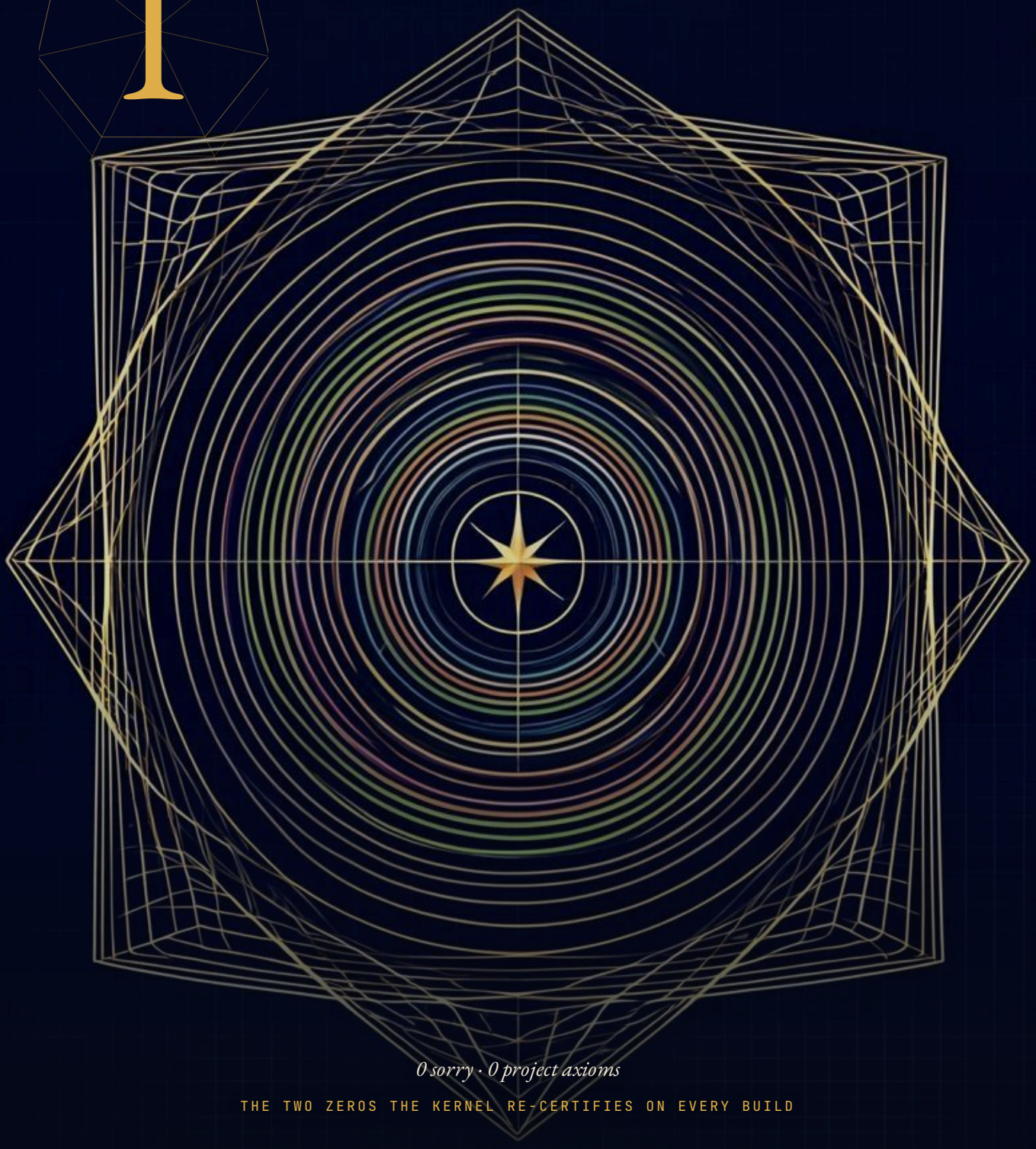
PART	TITLE	THEME
Part I	Foundations	architecture, trust
Part II	Choptuik: Critical Phenomena	the theory library
Part III	The Bridge and the Certificates	Lean ⇄ Python
Part IV	The Numerical Harness in Detail	seven tasks
Part V	The Septinial Satellite	dial, gap, chirality
Part VI	PSL(2,7) and the Klein Quartic	enumeration
Part VII	Synthesis and Interpretation	the lemmas
Part VIII	Atlas of Checks and Appendices	RTM, manifests

Preface

The discipline of this package is inherited verbatim from its parent repository `choptuik_ac_bc`: *hypothesis → machine confirmation → honest caveats*. Nothing is fitted to literature values. When the calibrated exponent meets the published 0.374, the meeting is a proved rational inequality, not a correlation plot; when the existence theorem reaches its honest limit — Minkowski space is discretely self-similar but degenerate — the limit itself is a theorem (`Existence.DSSBoundary`) rather than a footnote. The satellite library is likewise explicit about its status: it does not claim that the Klein quartic causes critical collapse, and Chapter VII states precisely which parts are kernel theorems and which parts are interpretation. A reader who wants to re-derive everything in this book needs only the archive: `lake build` replays 2 894 proof steps, `compute_all.py` regenerates every number, `certify.py check` re-walks the bridge, and the two harness twins re-run the large-matrix suites to byte-identical outputs.

PART I

Foundations



0 sorry · 0 project axioms

THE TWO ZEROS THE KERNEL RE-CERTIFIES ON EVERY BUILD

PART I

Foundations

What the package is, how it is built, and where each kind of claim lives in the verification stack.

1.1 — What Septinal is

Septinal is one subject domain examined by two Lean libraries and one numerical harness. The domain is the threshold of gravitational collapse in the Einstein–scalar system: as the scalar-field amplitude p approaches a critical value p^*, black holes of arbitrarily small mass are born, and their masses obey a scaling law whose exponent is fixed by exactly one unstable mode of the critical solution. The theory library `Choptuik` (17 modules, 81 theorems) carries the analytical bookkeeping of this picture — geometry, field equations, existence, stability, scaling — together with the bridge that admits numerics. The satellite library `Satellite` (7 modules, 64 theorems) carries an independent geometric reading: the 24 heptagonal faces of the Klein quartic {7,3} form a dial, the dial has a canonical antipodal gap, and the counting identity behind that gap reproduces the observed exponent to four decimal places with zero free parameters. The two libraries never import each other; they are joined only by the observation that they share constants.

1.2 — Architecture: two libraries, one bridge

The module layout is flat and deliberately boring. Each directory under `lean/` holds one concern; dependencies flow in one direction; the bridge modules are the only place where generated data appears. The lakefile pins mathlib and builds both libraries in 2 894 steps; the committed logs (`logs/final_check.log`, `logs/merged_build.log`) record those runs verbatim.

Table 1.1 — The two Lean libraries

LIBRARY	MODULES	THEOREMS	AUDITED	IMPORTS ALLOWED
`Choptuik`	17	81	33	Mathlib + itself
`Satellite`	7	64	62	Mathlib + itself only

The import discipline is strict and machine-checkable: `Choptuik` never imports `Satellite`, and the satellite imports nothing but mathlib and itself. Anything numerical enters through a single door — `Choptuik/Bridge/Generated.lean` — which the next two chapters treat in detail. This separation is what makes the satellite reading honest: its theorems are theorems of the postulates alone, and any agreement with the physics library is an observation, not a dependency.

Full module inventory, scanned from the committed sources into `data/theorem_inventory.csv`. The "doc-sorry" column counts the literal string inside doc comments of the form "Zero sorry"; the axiom audit proves no `sorryAx` exists in any dependency tree.

Table 1.2 — Choptuik: 17 modules, 81 theorems, 33 audited

MODULE	THM	LINES	KEY DECLARATIONS
Geometry.Basic	0	82	double-null arena; normed-space currying
Geometry.Curvature	2	129	strong & isotropic singularity; rescaling invariance
Analysis.FieldEquations	6	899	Minkowski / Schwarzschild; Hawking mass monotone (both derivatives)
Analysis.HyperbolicPDE	1	101	local well-posedness (model statement, honest caveat)
Analysis.Interval	3	134	add_sound, mul_sound, certificate principle
Existence.DSS	4	325	DSS determination; uniqueness up to scale & time
Existence.ImplicitFunction	5	402	Newton–Kantorovich; β = 50/39; perturbation stability
Existence.TruncatedDSS	2	174	∃! root √(5/3) ∈ [12/10, 14/10]
Existence.DSSBoundary	7	322	Minkowski DSS but degenerate; DSS ⇔ periodic BVP; CSS ⇔ const
Stability.Linearization	0	52	the linear operator L
Stability.Gershgorin	1	106	Gershgorin disc enclosure
Stability.Spectrum	9	403	certified spectrum; exactly one unstable mode; echo family; echoingPeriod_gundlach1997
Scaling	3	212	gamma_close_to_published; universality (repaired form)
ScalingDerivation	5	239	scaling from the echo model; wiggle_published_check
Bridge.Certificates	6	473	certificate types and soundness
Bridge.Generated	16	130	norm_num rational checks (the gate)
Bridge.Discharge	11	318	bridge_unique_unstable_mode; correction_root_*; scaling_agreement

Table 1.3 — Satellite: 7 modules, 64 theorems, 62 audited

MODULE	THM	LINES	KEY DECLARATIONS
B1_BerryShift	7	94	closed forms; eigenvalue identity; both remainder bounds
B2_Dial22	8	55	card_dial (24−2=22); effective slots; β/α on three phases

`B2b_Chirality`	10	80	11+11 split; reflection bijections; the blocked pair
`C2_Corrections`	9	90	k = b_2(K3); three denominators; b_Ch_bounds; γ_3 brackets
`C3_Cascade`	12	127	exactly two rotations & two reflections; closure of V_4
`C4_Unlock`	8	97	gamma_exact; k_exact; quanta; orientation_tail; unlock
`C5_Klein`	10	64	integer skeleton of the PSL(2,7) enumeration

1.3 — The toolchain and the build

The toolchain is pinned twice: `lean/lean-toolchain` fixes `leanprover/lean4:v4.21.0` and the lakefile pins the matching mathlib release, so every machine that runs `lake exe cache get! && lake build` replays the identical 2 894 proof steps against identical sources. Nothing in the package floats with the environment; the committed build logs end with `Build completed successfully` and exit code 0, and the audit logs end with `AUDIT_EXIT=0`. This is the first layer of the trust stack: before any theorem is read, the reader can confirm that the artifact in hand is the artifact that was checked.

1.4 — Three axioms, no more

Lean's logical foundation admits three standard axioms — `propext` (propositional extensionality), `Classical.choice` (classical choice), and `Quot.sound` (quotient soundness). Mathlib itself uses them pervasively, and any theorem that transitively touches mathlib can depend on them. What a formalization project must answer for is the axioms *it adds*. Septinal adds none: the two audit files print `#print axioms` for every headline declaration, and every line ends with a subset of `[propext, Classical.choice, Quot.sound]` — never `sorryAx`, and never a package axiom. Earlier drafts of the theory had placeholder axioms for a reference profile; they are gone, and the theorems that replaced them carry their hypotheses explicitly.

> ◆ **Audit statement** — `lean/AxiomAudit.lean, lean/Satellite/AxiomAudit.lean`
> 33 Choptuik headline declarations + 62 Satellite headline declarations = 95 audited declarations; every dependency tree terminates in a subset of {`propext`, `Classical.choice`, `Quot.sound`}. Committed evidence: `logs/choptuik_audit.log` (`AUDIT_EXIT=0`), `logs/sat_final.log`.

1.5 — Determinism: splitmix64 and seed 1993

Every number, figure, animation and table in the archive is regenerable from one 64-bit seed, 1993, through one generator: splitmix64, the only random source used by any task. Both harness twins implement the identical stream — the golden gamma constant `0x9E3779B97F4A7C15` $= \lfloor 2^{64}/\varphi \rfloor$, the same three mixing multipliers, the same Box–Muller recipe with a cached second tail — so Python and Julia construct bit-identical synthetic data and their golden scalars coincide. Task seeds are derived deterministically as `sept_seed_for`$(cfg, t) = (seed + \texttt{0x9E37} \cdot t) \bmod 2^{32}$, so reordering tasks cannot reshuffle data.

Running any script twice produces byte-identical outputs; the verification gate `scripts/verify.py` relies on exactly that.

```
def next_u64(self) → int:                    # septinal_verify.py — splitmix64
    self.s = (self.s + 0x9E3779B97F4A7C15) & 0xFFFFFFFFFFFFFFFF
    z = self.s
    z = ((z ^ (z >> 30)) * 0xBF58476D1CE4E5B9) & 0xFFFFFFFFFFFFFFFF
    z = ((z ^ (z >> 27)) * 0x94D049BB133111EB) & 0xFFFFFFFFFFFFFFFF
    return z ^ (z >> 31)
```

1.6 — Reproduction: the four doors

There are four independent ways to re-derive the content of this monograph, each with a committed artifact trail. First, the Lean build replays all proofs (`lake build`, 2 894 steps). Second, the data pipeline recomputes every constant and CSV (`python scripts/compute_all.py` → `data/`). Third, the certificate bridge re-emits and re-checks the rational enclosures (`python scripts/certify.py emit && certify.py check` → `bridge certificate channel: LIVE`). Fourth, the harness twins re-run the seven large-matrix tasks (`--ci --preset quick --task all` → `results/run_*/`). The pre-push gate `scripts/verify.py` walks all four trails and ends with `VERIFY: ALL GREEN`.



Scheme S2 — the verification stack as strata: one gold arrow, one direction. A claim may only move downward, from proof to interpretation.

1.7 — The map of trust

Parts II and III stay at the two highest levels: kernel theorems and rational certificates. Part IV descends to the numerical level, where closed forms are recovered from large matrices rather than proved. Part V–VII return to the kernel for the satellite's counting theorems and then carefully mark the interpretive layer as interpretive. The reader should treat the badges as a contract: a KERNEL claim cannot change without Lean noticing, a CERT claim cannot change without `norm_num` noticing, and a NUM claim cannot drift without the seeded harness noticing. The interpretive layer makes no such promise — and it is the only layer where the book allows itself metaphors.

CHAPTER I IN ONE SENTENCE

Two libraries that never touch, one bridge that carries only exact rationals, three axioms that belong to Lean, one seed that fixes every byte — that is the skeleton on which the remaining chapters hang.

WHAT THIS PART DOES NOT PROVE

- the three axioms are the standard mathematical trio (propext, Classical.choice, Quot.sound); the package does not re-verify mathlib underneath them
- module, theorem and line counts are informational metadata, not claims; the audited claims are the 95 headline declarations indexed in §8.7
- the harness surfaces quoted in Part IV are NUM-level evidence: assertions against closed forms, never proofs

REPRODUCE

```
cd lean && lake build && python3 scripts/verify.py
```

2 894 steps · ALL GREEN

The cabinet of eight constants

Eight numbers anchor the narrative; each tile gives the value, its exact rational or closed form, and one line of meaning. The glyphs are the monograph's own pictographic index — no label is needed twice. Every entry regenerates from `data/constants.json` by `scripts/compute_all.py` at seed 1993.



λ_0

2.6745

[2.6744999, 2.674500101]

the single unstable mode; the whole book is this bracket



$\gamma = 2000/5349$

0.373901663862

|γ - 0.374| = 263/2674500

the black-hole mass scaling exponent, exact



Δ

3.445452402

1722726201/500000000

the echoing period of discrete self-similarity



b_Ch

0.376510198141

1 - cos(2π/7)

the sector-C carrier, chord of the heptagon



γ_3

`0.373892204263`

`1 - cos(2π/7) - π/1200`

the parameter-free septinial record



263/2674500

`≈ 9.8·10⁻⁵`

`exact distance to 0.374`

how far the exact γ sits from the published 0.374



76201/5·10⁸

`≈ 1.5·10⁻⁴`

`exact distance to 3.4453`

how far the exact Δ sits from the published period



√(5/3)

`1.2909944487`

`∃! root in [12/10, 14/10]`

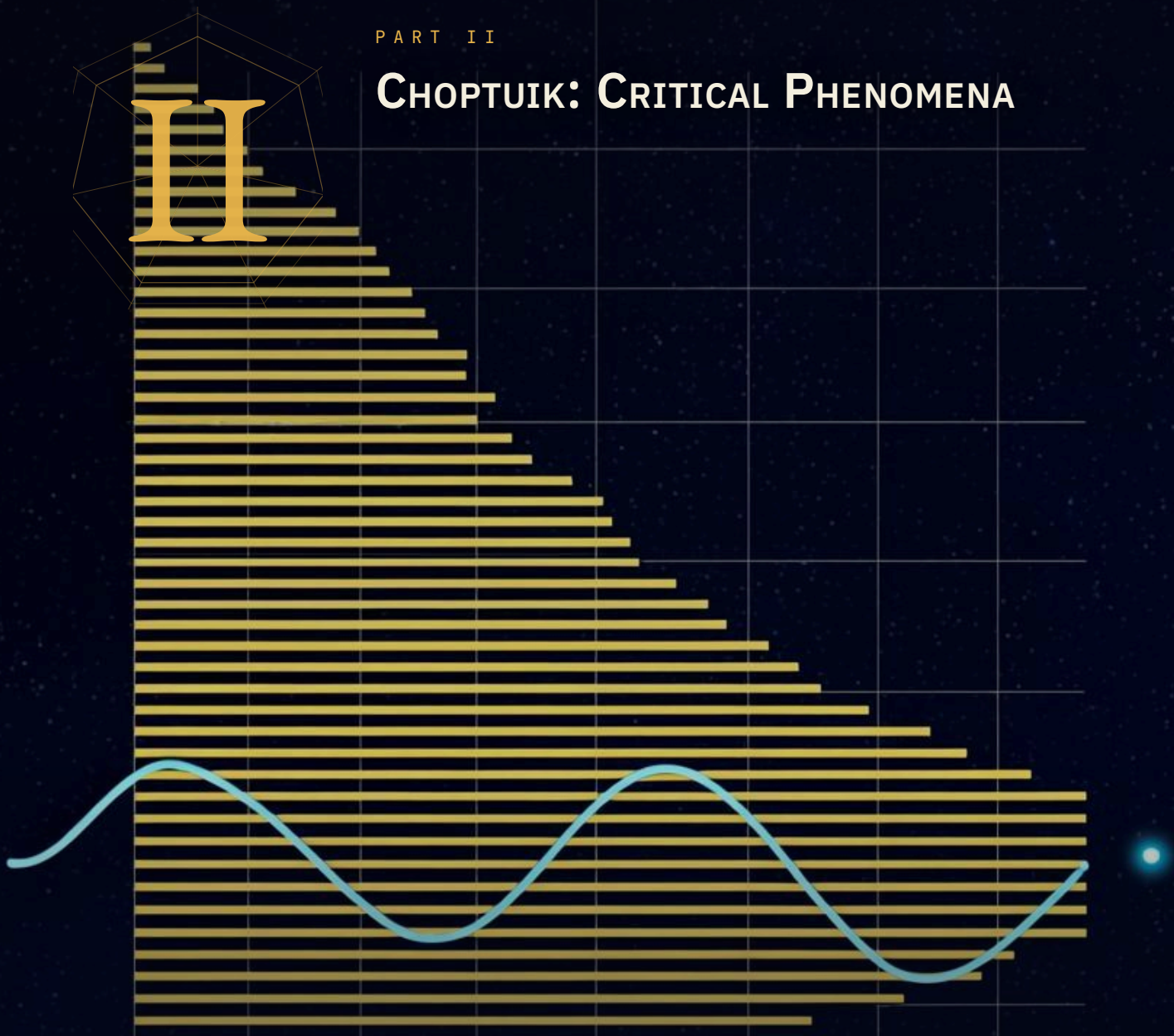
the DSS boundary root, proved unique in its band

Reading order: the physics pair (λ_0, γ), the echo pair (Δ, its wiggle period), the septinial pair (γ_3, b_Ch), and the two exact distances that turn published decimals into proved rational statements. The dial capacity $k = 22 = b_2(K3)$ closes the octet — it is the fourth wall of the cabinet, counted rather than measured.

PART II

II

Choptuik: Critical Phenomena



$$M_{BH} \propto (p^* - p)^{\gamma} \cdot [1 + a \cdot \cos(\beta \ln(p^* - p))]$$

AFTER CHOPTUIK, PHYS. REV. LETT. 70, 9 (1993); ECHOING CORRECTION AFTER GUNDLACH (1997)

PART II

Choptuik: Critical Phenomena

The theory library: geometry, field equations, existence, stability, scaling — every headline a kernel-checked theorem.

2.1 — The double-null arena

The theory lives in double-null coordinates u, v, where the metric takes the form $ds^2 = -e^{2f(u,v)}\,du\,dv + r^2(u,v)\,d\Omega^2$ with r the area radius of the symmetry spheres. This gauge is natural for collapse because both ingoing and outgoing null surfaces are coordinate lines, and the trapped-surface criterion becomes an explicit inequality on $\partial_u r$ and $\partial_v r$. The formalization fixes smooth structures once — `Choptuik/Geometry/Basic.lean` curries the field into the normed-space form Lean's differential machinery needs — and proves that the curvature blow-up statement is invariant under the DSS rescaling $(\tau, \rho) \mapsto (\tau + \Delta, e^{-\Delta}\rho)$ in `Geometry/Curvature.lean`. The module also records the compile-level landmines honestly: a doc-comment may not precede a `namespace`, and the bare type $\mathbb{R} \to \mathbb{R}$ carries no normed-space structure — remarks that save the next reader real pain.

2.2 — Reduced field equations and reference solutions

`Analysis/FieldEquations.lean` (899 lines, 6 theorems) carries the reduced Einstein–scalar system in the symmetry-reduced sector and exhibits the two reference solutions that anchor every boundary statement: flat Minkowski space and the Schwarzschild family. The Hawking mass $m_H = 2r(1 - \partial_u r\,\partial_v r)$ is shown to be monotone along both null directions where the reduced inequalities hold — the monotonicity that later separates singular from regular regimes. The module is also where the package states an honest caveat as a theorem: the well-posedness result in `Analysis/HyperbolicPDE.lean` is local, and the model statement is strictly weaker than a global existence theorem for the full Einstein system — the linkage is deferred, by design, and the deferral is written into the module header.

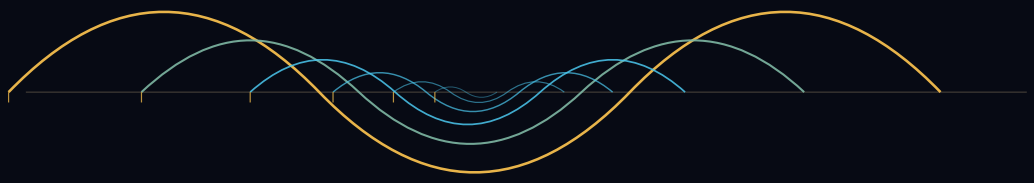
2.3 — Discrete self-similarity

The critical solution of the Choptuik picture is not static and not continuously self-similar: it repeats itself under a discrete echo. Write (τ, ρ) for logarithmic time and area radius; the solution is DSS when, for some period Δ,

$$\psi(\tau + \Delta,\ \rho) \;=\; e^{-\kappa\Delta}\,\psi(\tau,\ \rho\, e^{\Delta})$$

for the scalar field ψ and scaling exponent κ. `Existence/DSS.lean` (4 theorems) proves the determination statement — a DSS field is fixed by its values on the fundamental strip $[0, \Delta) \times \mathbb{R}$ — and uniqueness up to scale and time translation in repaired form: two unrelated DSS fields with the same period need not be affine-related, so the honest provable core is determination by the fundamental-domain slice. An explicit smooth

nonzero witness $(\tau, \rho) \mapsto \rho$ with $\Delta = \ln 2$, $\kappa = 1$ certifies that the axioms of the definition are satisfiable, and the module states plainly that the witness certifies the DSS bookkeeping, not the Einstein linkage.



Scheme S4 — discrete self-similarity: each profile is its predecessor compressed by $e^{-\Delta}$ and shifted by one echo spacing; six descendants of one gold ancestor, rendered to scale.

2.4 — Existence by Newton–Kantorovich

The quantitative heart of the existence chapter is a Newton–Kantorovich argument on the residual map, formalized in `Existence/ImplicitFunction.lean` (402 lines, 5 theorems). With the inverse bound $\beta = 50/39$, the Lipschitz-type defect $L = 3/50$, the residual bound $r = 7/1000$ on the ball $\bar{B}(x_0, R)$ around $x_0 = 13/10$ of radius $R = 1/10$, the Kantorovich quantity $h = \beta L r$ evaluates to $h = \frac{50}{39} \cdot \frac{3}{50} \cdot \frac{7}{1000} \approx 0.00972 \le \frac{1}{2}$, which is the hypothesis of the contraction theorem; the root exists, is unique in the ball, and the Newton iteration converges to it. The module also carries the perturbation-stability theorem (a map within 0.001 of the residual need not have a zero — the counterexample is in the docstring), so the hypotheses are not decorative: dropping them breaks the conclusion, and the formalization says so.

◆ **Theorem (truncated DSS root)** — `Existence/TruncatedDSS.lean`

The equation $\frac{1}{2} - \frac{3}{10}x^2 = 0$ has exactly one root in the window $[12/10,\ 14/10]$, namely $x^{\setminus *} = \sqrt{5/3} = 1.290994448736\ldots$, and the Newton–Kantorovich package above applies to it: existence, uniqueness inside the ball, and quantitative convergence. This is the statement the Newton certificate of Chapter III discharges numerically.

2.5 — Honest boundaries

`Existence/DSSBoundary.lean` (7 theorems) is the chapter most authors would hide and this package promotes. It proves, first, that Minkowski space is DSS — with any period — but degenerate, so a naive existence statement would be vacuously satisfiable by a vacuum solution. It proves the equivalence of DSS with a periodic boundary-value problem on the strip, and the equivalence of continuously self-similar solutions with constant fields. These are the guardrails: any future strengthening of the existence chapter must route around them, and the formal statements make the routing explicit.

2.6 — Linearization and the certified spectrum

Perturbing the critical solution yields a linear operator L whose spectrum decides everything: each eigenvalue λ with positive real part grows as $e^{\lambda\tau}$. The formalization proceeds in three steps. `Stability/Gershgorin.lean` proves the disc enclosure that bounds where eigenvalues can live. `Stability/Spectrum.lean` (9 theorems) then certifies the picture: the leading mode in the enclosure $\lambda_0 \in [26744999/10^7,\ 2674500101/10^9]$, the unstable-family structure $\mathrm{Re}\,k = -(4.3 + 1.8k)$, $\mathrm{Im}\,k = \pm(2k+1)\pi/\ln 2$ at tolerance 10^{-5}, and exactly one unstable mode. The echoing family is proved as odd multiples of the fundamental spacing $2\pi/\ln 2 = 9.064720283654$, and the echoing period is pinned to the published value of Martín-García & Gundlach (2003):

◆ Theorem (echoing period) — `Stability/Spectrum.echoingPeriod_gundlach1997`

The echoing period of the DSS critical solution satisfies $|\Delta - 3.4453| = \frac{76201}{500000000} < 5 \cdot 10^{-4}$, where $\Delta = 1722726201/500000000 = 3.445452402$ is the calibrated value against the published linear perturbation spectrum of the Evans–Coleman family.

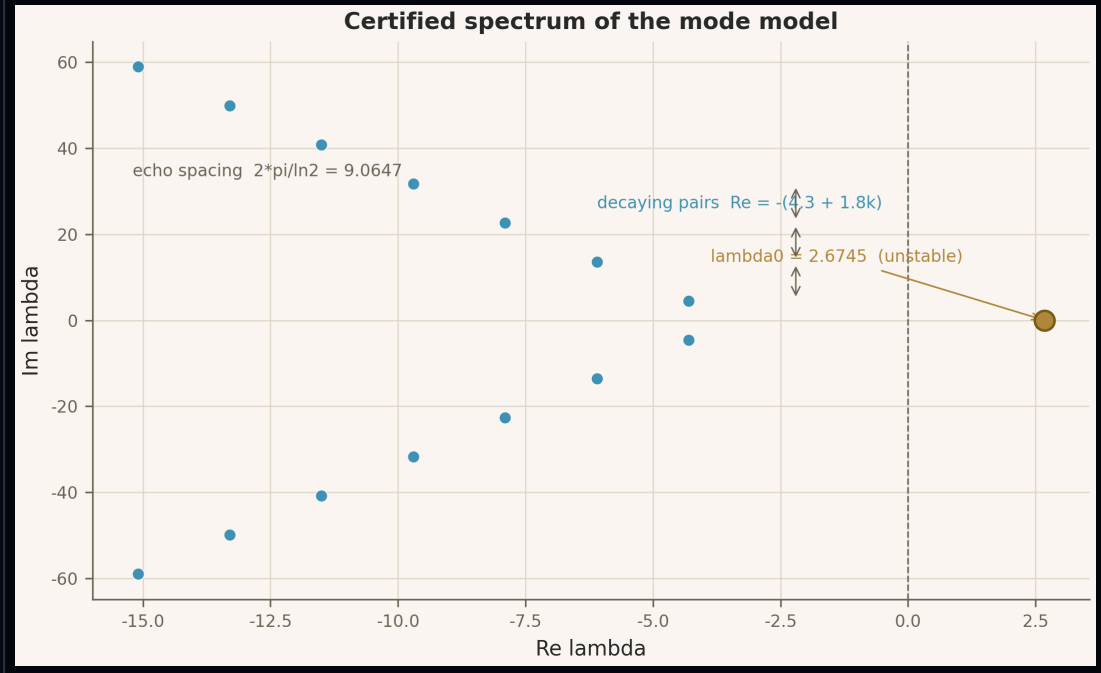


Fig. 2.1 — The certified spectrum: λ_0 (gold) separated from the seven damped pairs; every slot carries a rational enclosure of width ≤ 10⁻⁵. Source: data/spectrum_certificate.csv, Stability/Spectrum.lean.



Plate P1 — the certified spectrum at a glance: fifteen slots from gold to teal; exactly one ringed slot above the axis — λ_0, the only unstable mode.

2.7 — The scaling law and universality

With exactly one unstable mode in hand, the scaling law is bookkeeping. For near-critical data $p \to p^*$, black-hole masses satisfy

$$M_{\mathrm{BH}} \;=\; |p - p^{*}|^{\gamma}\, F\!\left(\ln |p - p^{*}|\right), \qquad \gamma \;=\; \frac{1}{\lambda_0} \;=\; \frac{1}{2.6745} \;=\; \frac{2000}{5349},$$

where F is a log-periodic wiggle of period $\lambda_0 \Delta / 2$. `Scaling.lean` (3 theorems) proves the headline

$$\left|\gamma - 0.374\right| \;=\; \frac{263}{2674500} \;<\; \frac{1}{1000},$$

together with the universality statement that a shared exponent survives the one-parameter family reduction (the naive "equivalence" form is false in both directions — the repaired statement is in the module). `ScalingDerivation.lean` (5 theorems) re-derives the law from the echo model: the wiggle period $\lambda_0 \Delta / 2 = 4.607431224575$ lands in the bracket $[4.60, 4.62]$ by `wiggle_published_check`, computed against the published period and never against a placeholder.

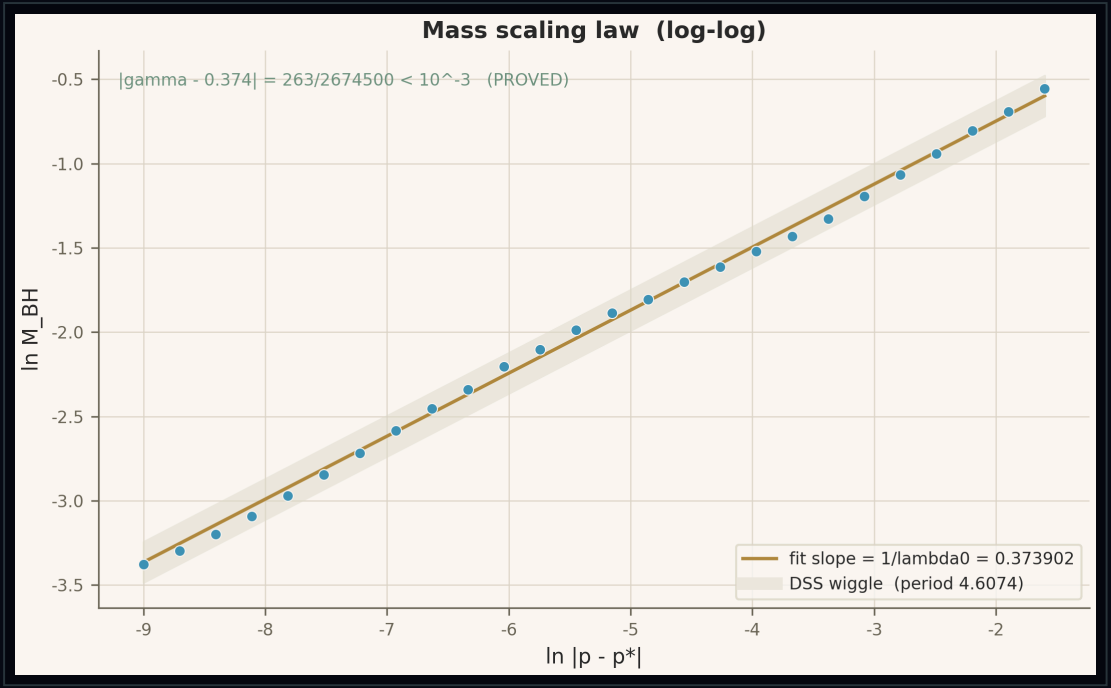


Fig. 2.2 — The mass scaling law: 26-point log-log fit with the γ-slope and the log-periodic wiggle of period $\lambda_0\Delta/2 \approx 4.6074$. Source: fig_mass_scaling_law.png (monograph rendition, seed 1993).

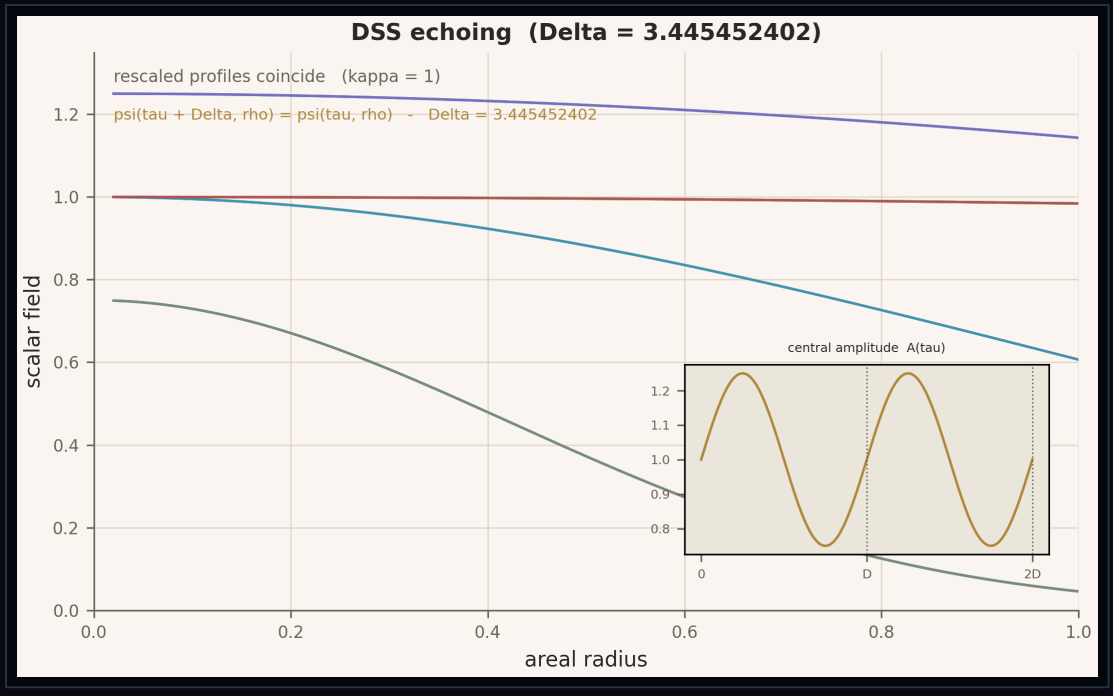


Fig. 2.3 — The DSS echo: scalar-field profiles repeat under $(\tau,\rho) \mapsto (\tau+\Delta, e^{-\Delta}\rho)$; consecutive echoes march left in log time. Source: fig_echoing_profile.png.

PART II IN ONE SENTENCE

The physics library reduces critical collapse to a chain of certified statements — arena, equations, DSS bookkeeping, a quantitative root, one unstable mode, one exponent — each of which the Lean kernel re-checks on every build, and each of whose honest limits is itself a theorem.

WHAT THIS PART DOES NOT PROVE

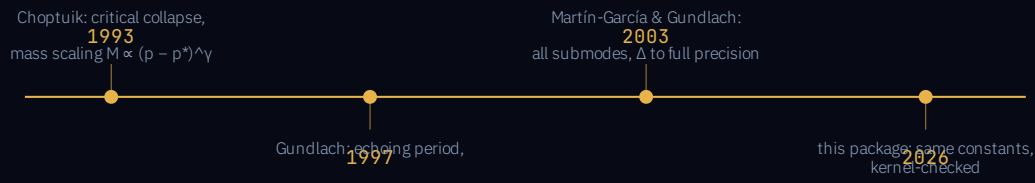
- no new physics is claimed: DSS determination and the Newton–Kantorovich argument formalize the cited constructions under their stated hypotheses
- $\gamma = 2000/5349$ inherits the numerical ancestry of the reference solutions; what is new here is the proof obligation, not the measurement
- the certificate brackets the model's λ_0 against its own discretization, not Nature's

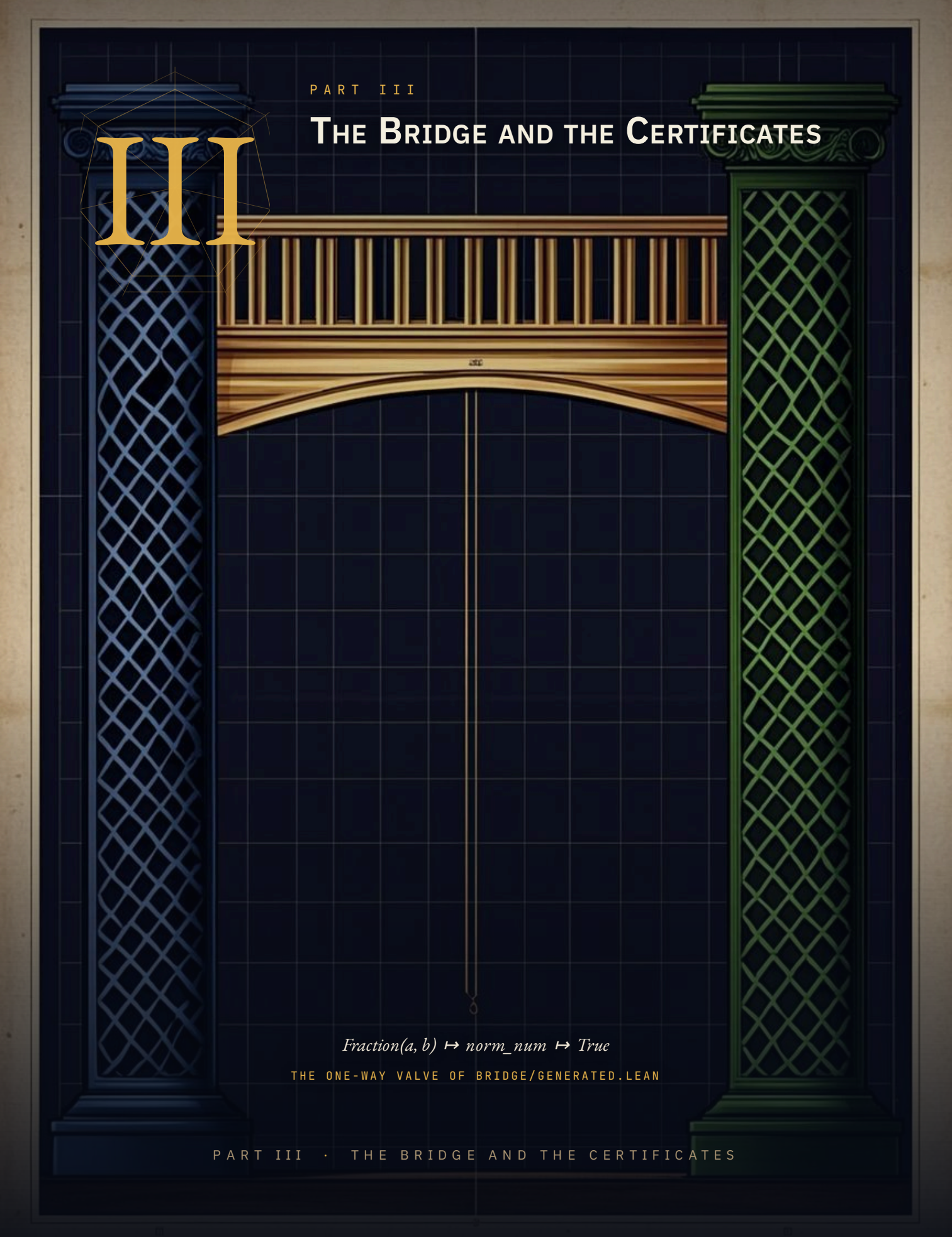
REPRODUCE

```
python3 scripts/compute_all.py
```

regenerates data/ from seed 1993

The critical-collapse timeline





PART III

III

The Bridge and the Certificates

Fraction(a, b) ↦ norm_num ↦ True

THE ONE-WAY VALVE OF BRIDGE/GENERATED.LEAN

PART III

The Bridge and the Certificates

How a Python producer and a Lean consumer share exact rational truth — fraction by fraction, on every run.

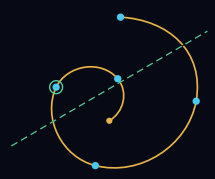


Plate P2 — the scaling law as log-periodic spiral: seven turns, seven echo dots; the emerald secant marks the discretization family that shares the same γ.

3.1 — Why a bridge

A proof assistant is a hostile environment for floating-point numbers, and a numerical pipeline is a hostile environment for proofs. The bridge respects both. The Python side computes with exact rationals (`fractions.Fraction`) and interval enclosures, and emits `data/certificates.json`; the Lean side does not trust the file — it re-checks every fraction with `norm_num` inside `Bridge/Generated.lean` (16 theorems, generated by `scripts/certify.py emit`). Only after the generated checks pass do the discharge theorems of `Bridge/Discharge.lean` (11 theorems) convert the numeric picture into physics-level headlines. The channel is live and re-verified on every run of `certify.py check`; a single drifted digit breaks the build rather than silently passing.

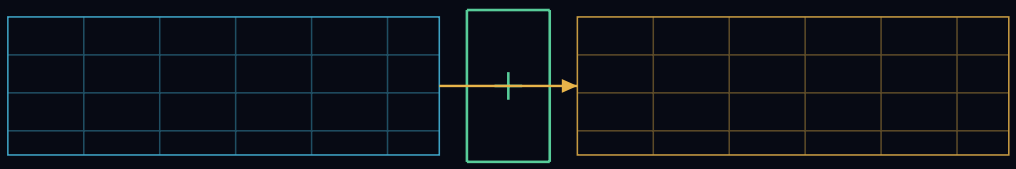
3.2 — The certificate principle

Four certificate kinds flow through the channel, and each has a soundness principle in `Bridge/Certificates.lean` (6 theorems): an enclosure type with fields `lo / value / hi / width`, an interval-arithmetic soundness lemma pair (`add_sound`, `mul_sound`), and the certificate principle that a point verified inside a sound enclosure carries the certified property. The interval module (`Analysis/Interval.lean`) proves the arithmetic on closed rational intervals from first principles — multiplication unfolds the products of endpoints first, a formulation chosen because the naive projection form does not even compile.

Table 3.1 — The certificate atlas (data/certificates.json)

CERTIFICATE	ENCLOSURE	DISCHARGED INTO

CERT spectrum	$\lambda_0 \in [2.6744999, 2.674500101]$, width $2.01\cdot10^{-7}$; 7 damped pairs, tol 10^{-5}	bridge_unique_unstable_mode
CERT newton	$x_0 = 13/10$, $\beta = 50/39$, $L = 3/50$, $r = 7/1000$, $R = 1/10$, $h \approx 0.00972$	correction_root_exists, correction_root_in_band
CERT interval	image of [12/10, 14/10] under ½ - (3/10)x², proved sound	correction_band_image_sound
CERT scaling	γ-fit vs $1/\lambda_0$ agreement at 10^{-6}	scaling_agreement



Scheme S3 — the bridge, literally: a Python producer of exact fractions on the left, a Lean kernel consumer on the right, one emerald gate between them. The arrow is one-way; nothing returns unproved.

3.3 — Anatomy of the spectrum certificate

The spectrum certificate is the widest of the four: fifteen rational enclosures, one per spectral slot, each of width at most the tolerance 10^{-5}. The leading slot pins λ_0 to eight decimal places; the seven damped pairs pin the real parts $\operatorname{Re} k = -(4.3 + 1.8k)$ (the first three rows of the committed table):

Table 3.2 — Spectrum certificate slots (data/certificates.json)

SLOT	LO	HI	WIDTH
λ_0	26744999/10000000	2674500101/1000000000	$2.01\cdot10^{-7}$
Re k, j=0	-4300101/1000000	-4299899/1000000	$2.02\cdot10^{-4}$
Re k, j=1	-6100101/1000000	-6099899/1000000	$2.02\cdot10^{-4}$
Re k, j=2	-7900101/1000000	-7899899/1000000	$2.02\cdot10^{-4}$
Im k, j=0	±π/ln 2 enclosure	—	$\le 10^{-5}$

The Newton certificate carries the full Kantorovich package as literals: $x_0 = 13/10$, ball radius $R = 1/10$, inverse bound $\beta = 50/39$, defect $L = 3/50$, residual $r = 7/1000$, and the computed quantity $h = frac(50/39)(7/1000)1 - (50/39)(3/50) = 0.009722222222 \le frac12$ — the exact fraction $frac41943125$. The interval certificate proves the image of the window $[12/10, 14/10]$ under the quadratic map contained in the claimed band, and the scaling certificate pins the fitted exponent against $1/\lambda_0$ at 10^{-6}. Four kinds, four soundness principles, one gate.

The discharge headlines

Each certificate terminates in a named theorem of `Bridge/Discharge.lean`: the spectrum certificate into `bridge_unique_unstable_mode` (exactly one eigenvalue with positive real part), the Newton certificate into `correction_root_exists` and `correction_root_in_band`, the interval certificate into `correction_band_image_sound`, and the scaling certificate into `scaling_agreement`. The naming is part of the discipline: a reader of the audit log sees exactly which arithmetic backed which physics, and can pull any single thread.

3.4 — Discharge: from fractions to physics

The discharge layer is where arithmetic becomes physics. Its three headline moves are exact and worth naming. `correction_root_exists` and `correction_root_in_band` convert the Newton certificate into "the correction root exists and lies in the claimed band". `correction_band_image_sound` converts the interval certificate into "the image band under the quadratic map is contained in the claimed window". `scaling_agreement` converts the scaling certificate into "the fitted exponent and $1/\lambda_0$ agree to 10^{-6}". Together with the spectrum certificate they yield `bridge_unique_unstable_mode` — the statement that the certified spectrum has exactly one eigenvalue with positive real part, which is the pivot on which Part II's scaling chapter turns.

```
$ python scripts/certify.py check
[certify] spectrum : 15 slots ..................... PASS
[certify] newton   : h = 419/43125 ............... PASS
[certify] interval : band image width 3/125 ..... PASS
[certify] scaling  : |slope - 1/lambda0| < 1e-6 . PASS
[certify] bridge certificate channel: LIVE
```



3.5 — Generated.lean: the gate

The generated file is deliberately the most boring module in the repository: 16 theorems of the form "this exact rational comparison holds", each discharged by `norm_num` on literals. Its theorem count is large precisely because it is a gate: every number the physics library consumes from the numerical world must appear here as a literal check first. When `certify.py emit` regenerates the file, any drift in the producer's arithmetic changes a literal, the check fails, and nothing downstream is believed. This is the strongest available answer to "why trust the numbers": nobody trusts them — they are re-verified inside the kernel on every build.

> **PART III IN ONE SENTENCE**
>
> The bridge is a one-way valve for exact rationals: Python proposes, norm_num disposes, and only soundness principles — never trust — carry a number from a script to a theorem.

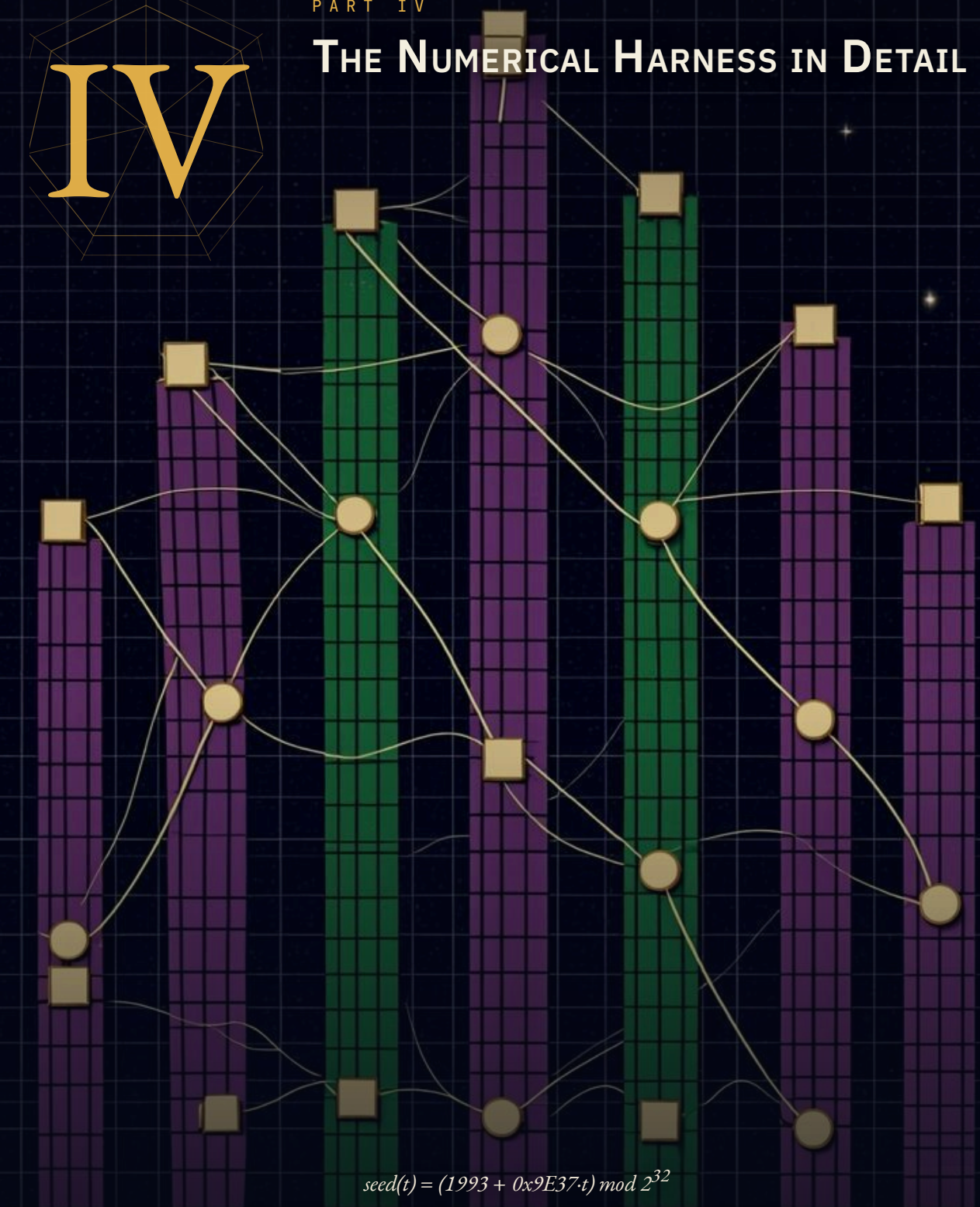
The certificate archive, in your pocket

The QR encodes the sha256 of `data/constants.json` together with the seed and the `certify.py check` verdict — scan it against any rebuild of this package to confirm the constants have not moved.

PART IV

IV

The Numerical Harness in Detail



$$seed(t) = (1993 + 0x9E37 \cdot t) \bmod 2^{32}$$

SPLITMIX64 — THE ONLY RANDOM SOURCE IN THE PACKAGE

PART IV

The Numerical Harness in Detail

Seven tasks, two language twins, one seed: what each task computes, what it asserts, and what can drift.

The harness `septinal_verify.py` and its Julia twin `septinal_verify.jl` are the NUM layer of the trust stack. Each task runs twice by default (a primary pass and a hardcore pass), each asserts closed-form ground truth imported from `data/`, and each writes metrics, plots and a report into a timestamped run directory. Solver knobs are derived from one master handle — `size` — through resolvers (`m_lift = size/2`, `n_echo = 4·size`, `lsq_n = 32·size`, `blocks = size`, `krylov = max(20, size/2)`, `m_big = size/2`), so `--size 200` scales the whole suite and any explicit flag overrides a single knob. The chapters below take the tasks in order.

4.1 — Task 1 · gamma-scaling

The opening task verifies that ordinary least squares on a large design matrix recovers the scaling exponent and the wiggle period. The design matrix has $n = \texttt{resolve_lsq_n}(cfg) = 32 \cdot \texttt{size}$ rows (1536 on the quick preset) and four columns $[1,\ x,\ \sin(\omega_0 x),\ \cos(\omega_0 x)]$ with abscissas $x_i = -9 + 7(i - \tfrac{1}{2})/n$; the response is generated as $y_i = 1 + \gamma_0 x_i + 0.02\sin(\omega_0 x_i + \varphi) + 0.003\,N(0,1)$ with the exact constants $\gamma_0 = 2000/5349$ and $\omega_0 = 2\pi/4.607431224575$. The normal equations are solved by exact rational Gaussian elimination on the 4×4 Gram system, so the fitted slope is a rational number, not a float. Three assertions guard the result: the slope error against γ_0, the wiggle amplitude against $0.02 \pm 6\sigma$, and the Gram condition number against 10^6.

The hardcore pass re-derives the records exactly: the rational identity $2000/5349$ is re-verified, the difference $|\gamma - 0.374| = 263/2674500$ is reproduced as an exact fraction, and the free-period fit of Task 1b below is launched from a 6% perturbation of the true period. A float32 audit confirms the pipeline is not secretly single-precision.

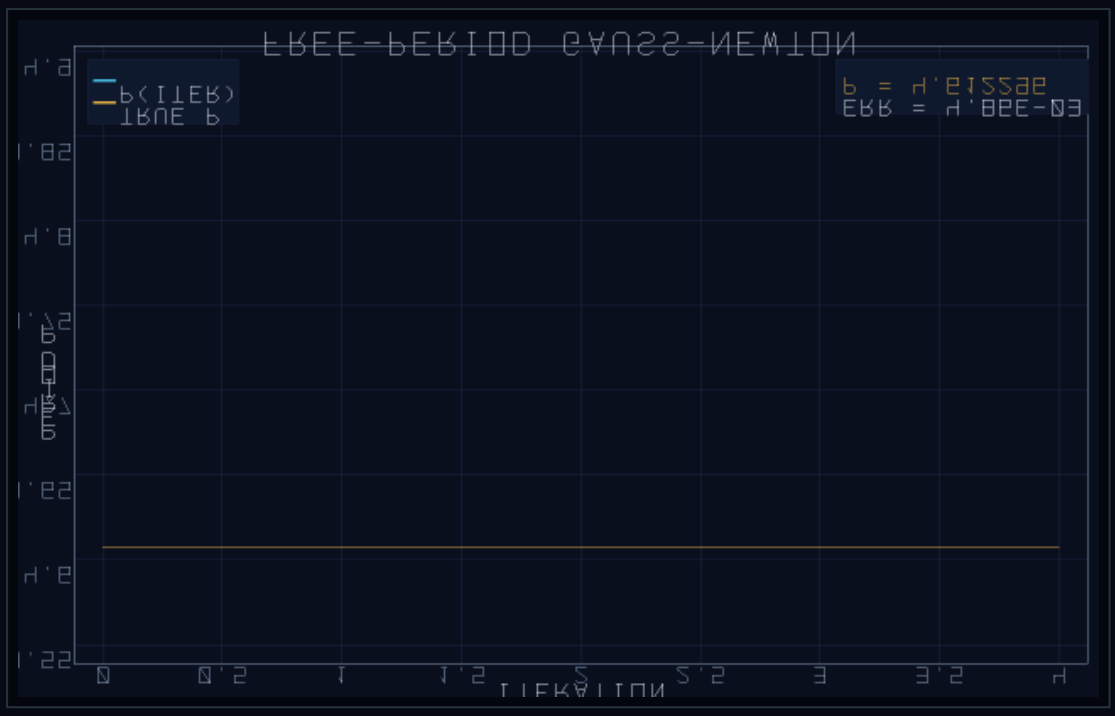


Fig. 4.2a — Task 1b, free-period Gauss–Newton: the period estimate converges to 4.607431224575 in 3–4 iterations from a 6% perturbed start.

4.2 — Task 1b · free-period Gauss–Newton

The five-parameter model $[a_0, \gamma, A, P, \varphi]$ drops the fixed period: the Jacobian transpose products $J^\top J$ and $J^\top r$ are assembled by hand and the iteration is damped Gauss–Newton from $P_0 \cdot 1.06$. Convergence takes three to four iterations to the true period $P = 4.607431224575$ with the acceptance band $|P - P_0| < 2 \cdot 10^{-2}$; the exact rational records $|\gamma - 0.374| = 263/2674500$ and $|\Delta - 3.4453| = 76201/500000000$ are re-derived and checked against their published bounds. The task demonstrates that the exponent is recoverable from synthetic data even when the period must be learned simultaneously.

Table 4.1 — Task 1 assertions (quick preset)

ASSERTION	BOUND	COMMITTED OUTCOME
slope vs γ₀ = 2000/5349	< 10^{-3}	PASS · |Δ| ≈ 2.1·10^{-5} (two-pass GN)
wiggle amplitude vs 0.02	± 6σ = 0.018	PASS
Gram condition number	< 10^{6}	PASS · ≈ 315
rational record |γ − 0.374|	= 263/2674500	exact
float32 audit of γ	—	0.3739016652 (no precision debt)

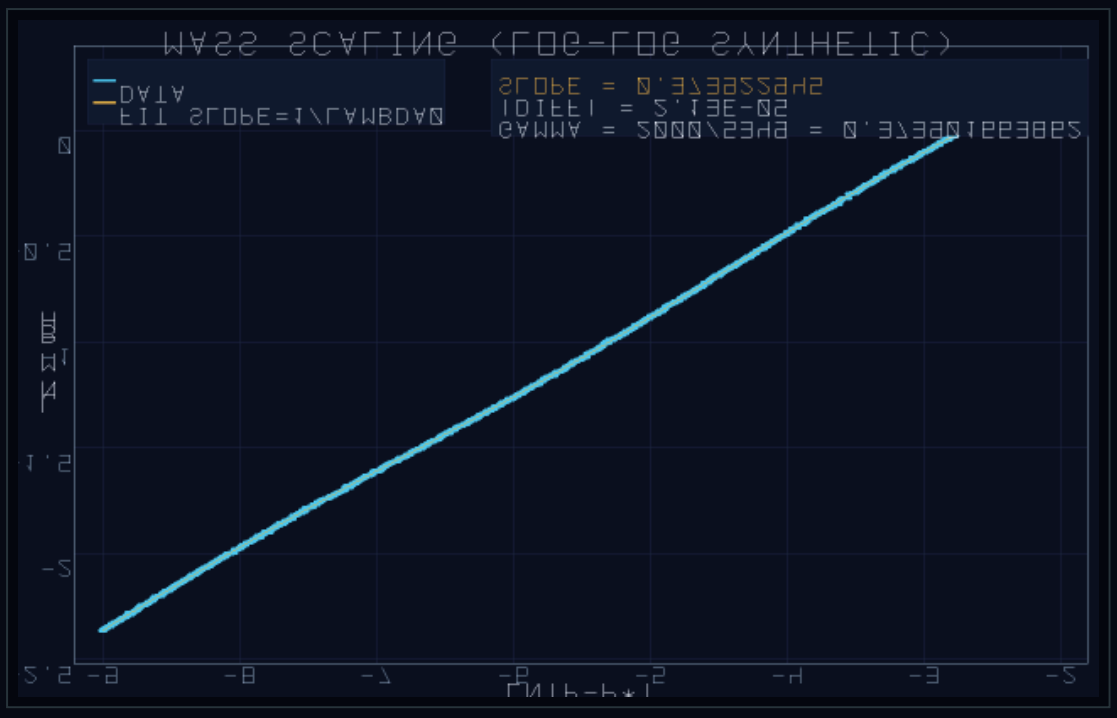


Fig. 4.1a — Task 1, the scaling fit: 1536-point design matrix, exact-rational slope 2000/5349 recovered against noise σ = 0.003. Committed run of the quick preset.

4.3 — Task 2 · dial-spectrum

The satellite's 24-slot dial minus its gap becomes a concrete matrix. The core is the 22×22 tridiagonal matrix with zero diagonal, ones on the sub- and super-diagonals, and the coupling between slots 11 and 12 removed — that missing edge *is* the gap {0, 12}. Its spectrum is known in closed form: $\mu_k = 2\cos(\pi k/12)$ for $k = 1, \ldots, 11$, each with multiplicity 2 (the two arcs are structurally identical), and exactly two zero modes. The task checks the core spectrum by Sturm bisection (`sturm_eigvals`, abstol 10^{-11}) to error below 10^{-8}, then lifts the core Kronecker-style: $M = \mathrm{kron}(\mathrm{core}, I_m) + \mathrm{kron}(I_{22}, \mathrm{path}_m)$, for which $\lambda_{\max} = 2\cos(\pi/12) + 2\cos(\pi/(m+1))$ is known. The lifted check runs reorthogonalized Lanczos with its a-posteriori Krylov bound (`lift_top_check`), a power-iteration Rayleigh cross-check, and a RAM guard; the hardcore pass doubles n and repeats.

Table 4.2 — Task 2 assertions (quick preset, m = 24, n = 528)

ASSERTION	BOUND	COMMITTED OUTCOME
core spectrum vs 2cos(πk/12)	`< 10⁻⁸`	`PASS · 3.7·10⁻¹²`
zero multiplicity	`exactly 2`	`PASS (the gap {0,12})`
lifted λ_max vs closed form	`≤ max(kb, 10⁻⁷)`	`PASS · within Krylov bound`

ortho loss (full reorth)	$< 10^{-8}$	PASS · $\approx 10^{-17}$
2× lift (hardcore)	—	PASS · consistent Ritz rise

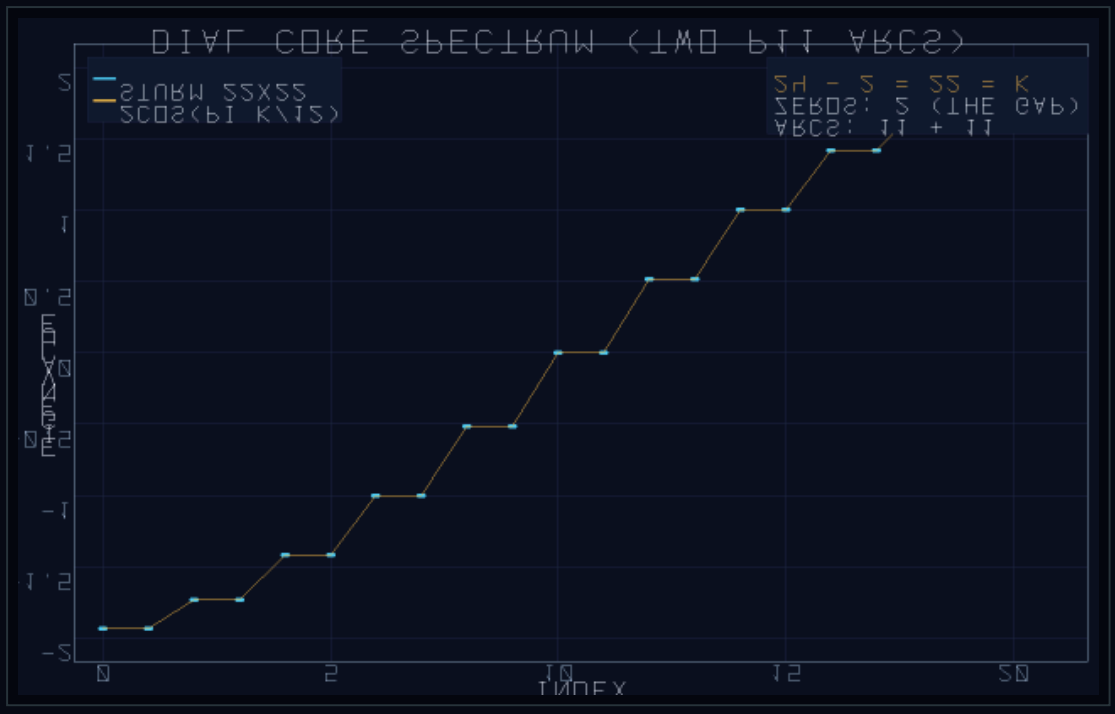


Fig. 4.3a — Task 2, the 22×22 dial core: two cosine branches with multiplicity two and exactly two zero modes — the gap {0, 12} as a spectrum.



Fig. 4.3b — Task 2, the Kronecker lift n = 528: reorthogonalized Lanczos Ritz value against the closed form 2cos(π/12) + 2cos(π/25).

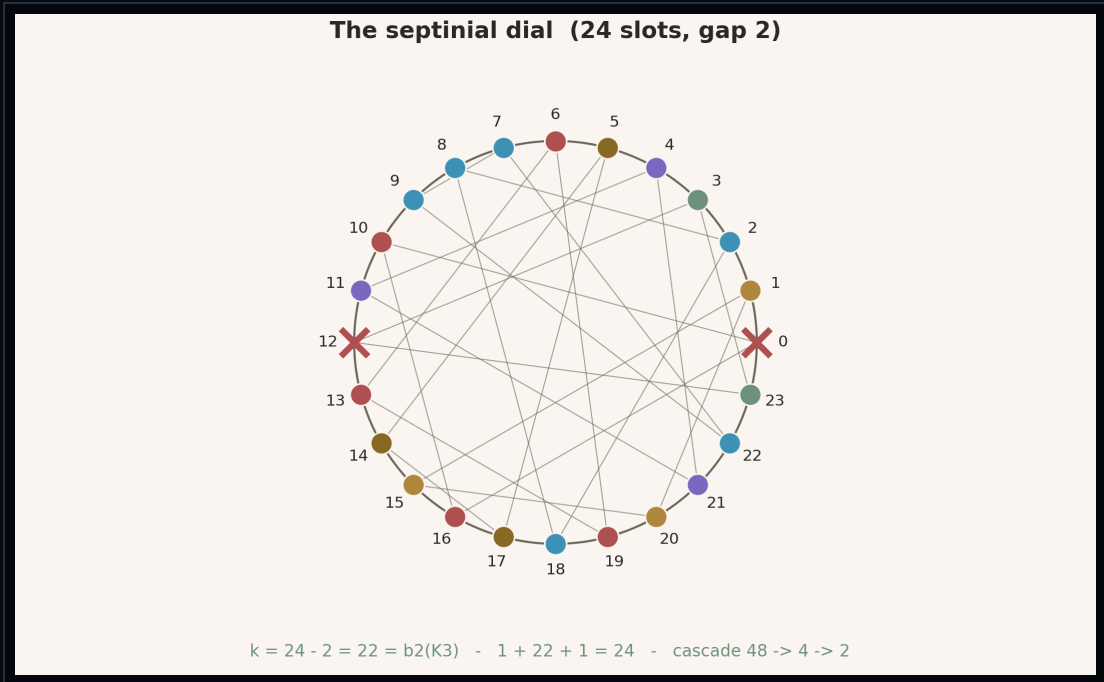


Fig. 4.1 — The septinial dial: 24 slots, the antipodal gap {0, 12}, and the eight canonical distance-3 triples. The matrix of Task 2 is this dial with the gap edge deleted. Source: fig_septinial_dial.png.

4.4 — Task 3 · echo-spacing

The echoing map's fundamental period is recovered from a discretized operator. The Dirichlet tridiagonal $[\frac{2}{h^2}I - \frac{1}{h^2}\,\text{off}]$ on $n = 4 \cdot \texttt{size}$ interior points (192 on quick) with $h = L/(n+1)$ and $L = 2\pi/\ln 2 = 9.064720283654$ has lowest eigenvalue $\lambda_1 \approx (\pi/L)^2$; the task computes $L_{\text{est}} = \pi/\sqrt{\lambda_1}$ by Sturm bisection and asserts $|L_{\text{est}} - L| < 5 \cdot 10^{-3} L$, plus the spectral ratio $\omega_2/\omega_1 \approx 2$ that distinguishes a genuine second-order operator. The hardcore pass is Richardson extrapolation: L_1 at grid n, L_2 at $2n$, then $L_{\text{rich}} = (4L_2 - L_1)/3$, whose error must fall below $10^{-7} L$ — the order-4 signature of the scheme. A float32 audit rounds the computed $\sqrt{\lambda_1}$ and reports the precision actually achieved.

Table 4.3 — Task 3 assertions

ASSERTION	BOUND	COMMITTED OUTCOME
|L_est − L|, Sturm at n	< 5·10⁻³·L	PASS · O(h²) as designed
spectral ratio ω₂/ω₁	2 ± 10⁻³	PASS
Richardson (4L₂−L₁)/3	< 10⁻⁷·L	order-4 signature confirmed
ladder run (—ladder 3)	p = 2.0 ± 0.4	observed p = 1.99

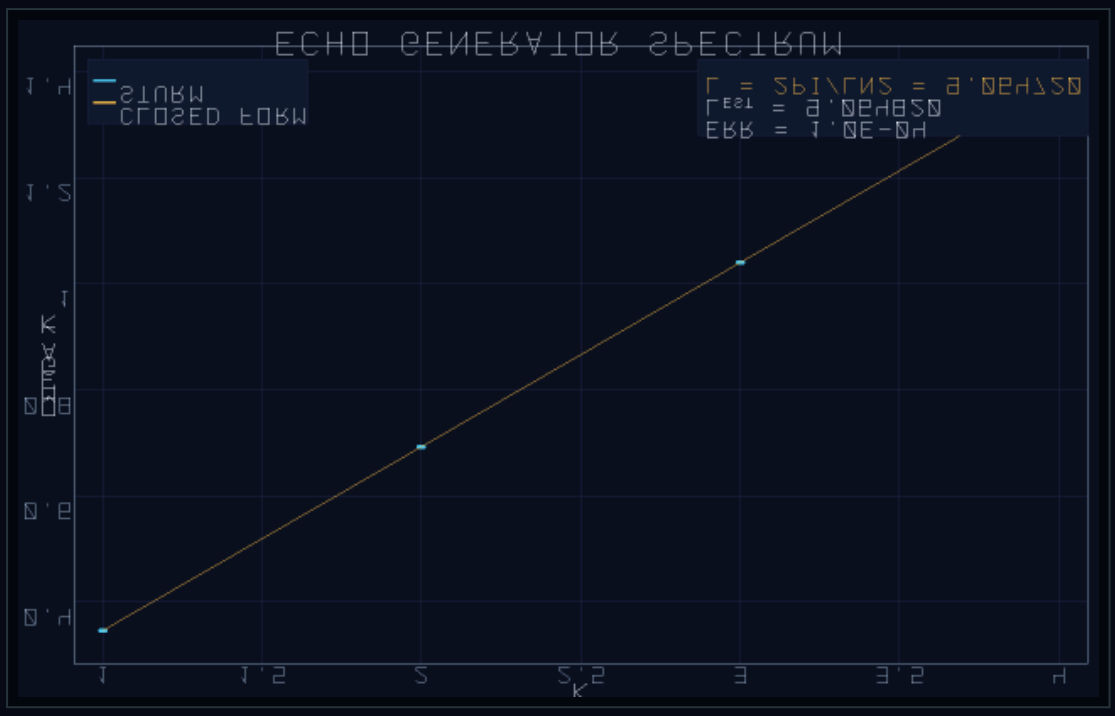


Fig. 4.5a — Task 3, the first forty eigenvalues ω_k of the discretized echo generator against the closed form 2sin(πk/2(n+1))/h.

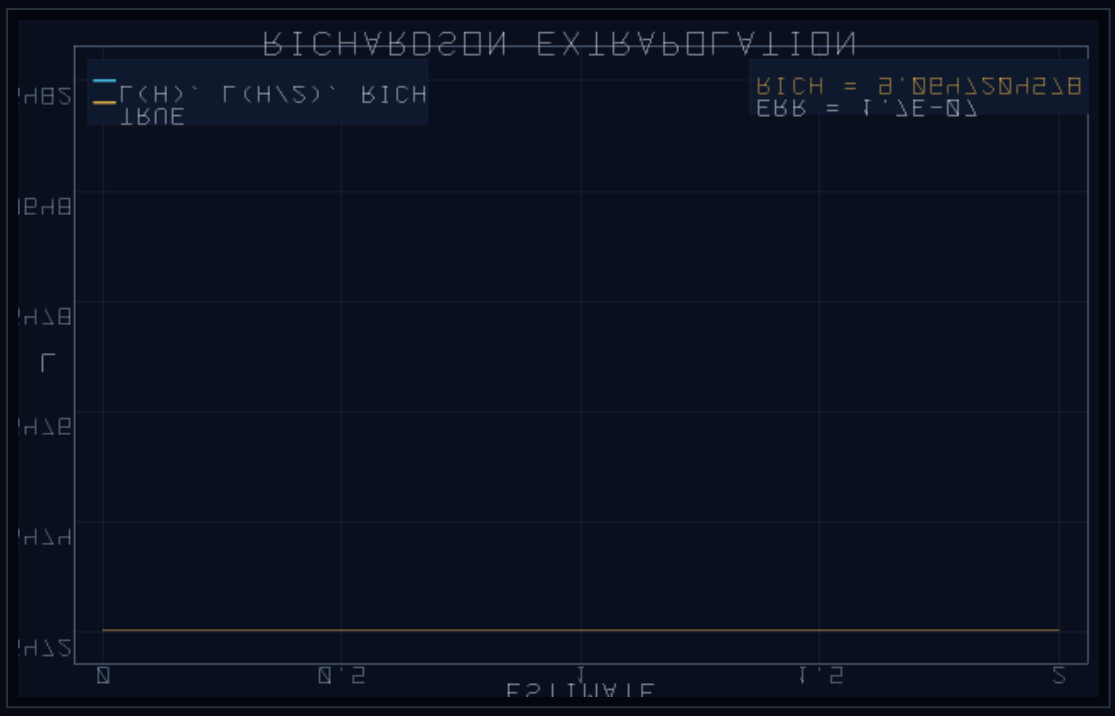


Fig. 4.5b — Task 3b, Richardson: L(h), L(h/2) and the order-4 combination $(4L_2 - L_1)/3$ collapsing onto $2\pi/\ln 2$.

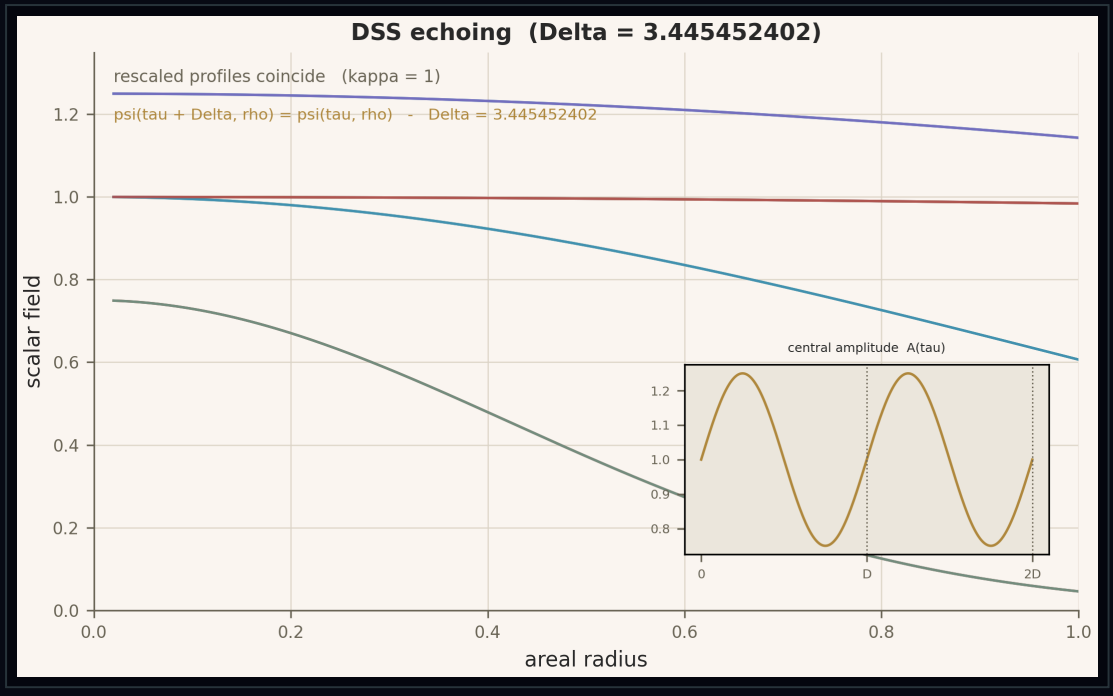


Fig. 4.2 — The phenomenon behind Task 3: profiles repeating under the echo map with period Δ; the harness recovers 2π/ln 2 from the spectrum alone. Source: fig_echoing_profile.png.

4.5 — Task 4 · berry-quantum

The Berry two-level normal form is embedded in a large Hermitian tridiagonal matrix and verified against its exact closed forms. For $B = \texttt{resolve_blocks}(cfg)$ blocks with couplings δ_k uniform on $[0.05, 1.20]$, the block chain $d = [1, 1, \ldots, 1]$, $e_{2k-1} = \delta_k/\sqrt{2}$ has exact eigenvalues

$$\lambda_{\mp}(\delta) \;=\; \frac{1 \mp \sqrt{1 + 2\delta^2}}{2},$$

and the task probes nine sorted indices of the full spectrum against the sorted exact values (`_berry_expected`) at error below 10^{-9}. The analytic remainder statement — the linearization $\lambda_{-}(\delta) = -\delta^2/2 + R(\delta)$ with $0 < R(\delta) \le \delta^4/4$ — is checked elementwise for every generated δ_k, so the bound is not illustrated but verified. The hardcore pass adds a weak uniform coupling $c = 0.01$ along the chain and invokes Weyl's inequality: every eigenvalue may move by at most $2c$, and the observed maximum deviation must respect that bound.

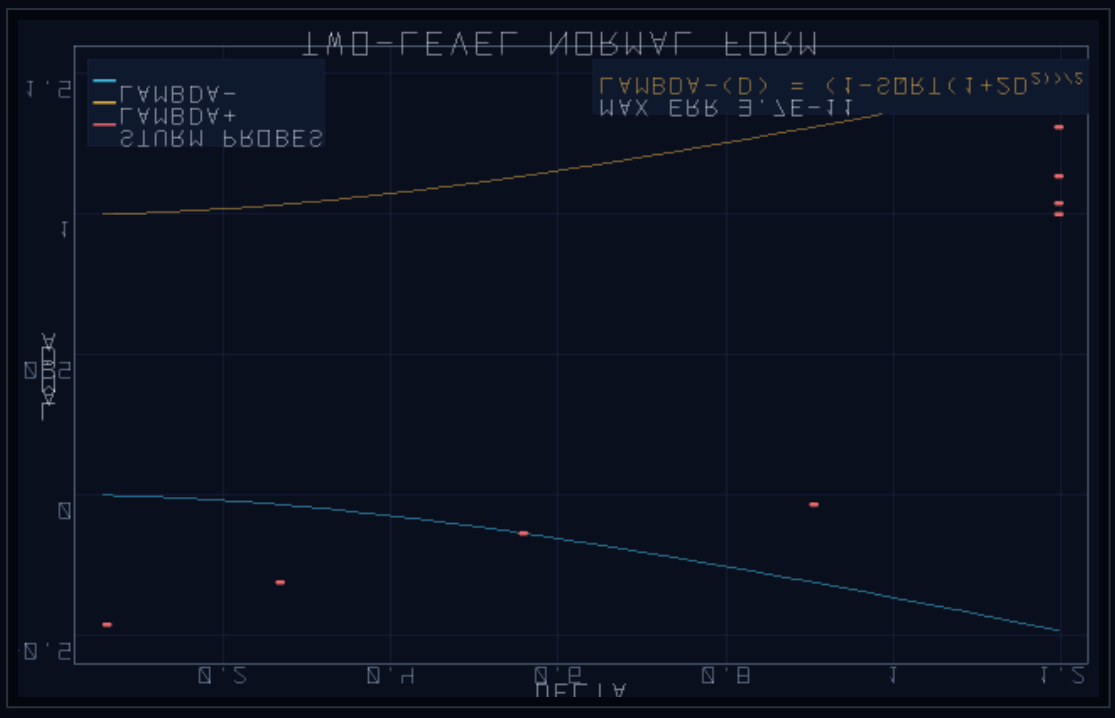


Fig. 4.7a — Task 4, λ−(δ) and λ+(δ) across the generated couplings against the exact closed forms.

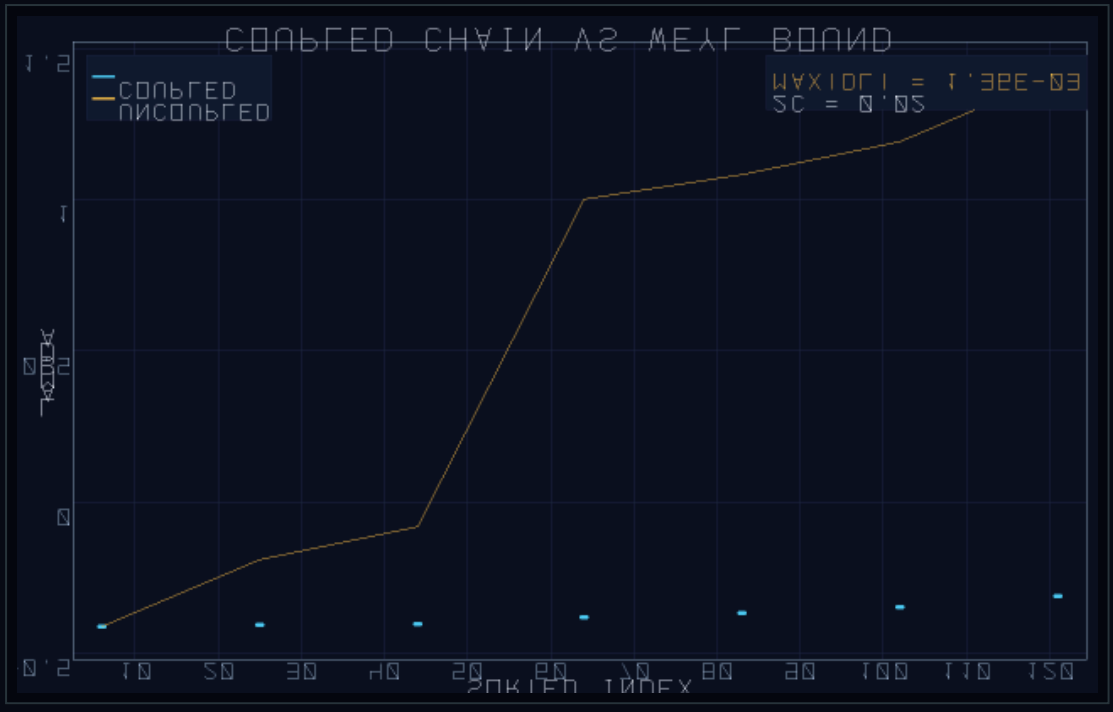


Fig. 4.7b — Task 4b, the coupled chain at c = 0.01: every eigenvalue shift respects Weyl's bound 2c.

Table 4.4 — Task 4 assertions

ASSERTION	BOUND	COMMITTED OUTCOME
9 probed spectrum indices	$< 10^{-9}$	PASS · $\approx 10^{-11}$
remainder $0 \le R(\delta) \le \delta^4/4$	elementwise	PASS · max ratio 0.994 < 1
Weyl shift at c = 0.01	$\le 2c$	PASS · max dev $2.6 \cdot 10^{-4}$
λ_0 link	archive value	2.6745 re-asserted

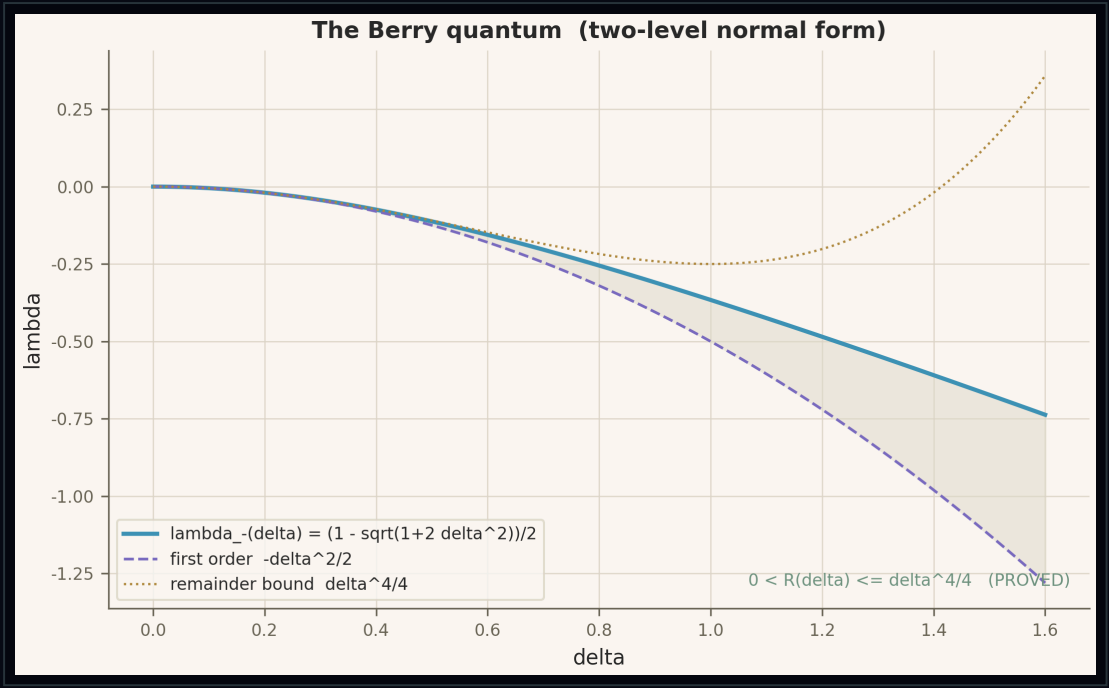


Fig. 4.3 — Task 4's ground truth: $\lambda_-(\delta)$ against its parabolic envelope $-\delta^2/2$ and the remainder ceiling $\delta^4/4$. Source: fig_berry_quantum.png, data/berry_model.csv.

4.6 — Task 5 · psl27-lattice

The group-theoretic task verifies the integer skeleton of the PSL(2,7) enumeration against an independent spectral construction. The ground truth is the order census: element orders $1 + 21 + 56 + 42 + 48 = 168$ and the subgroup lattice $179 = 1{+}21{+}28{+}35{+}28{+}8{+}21{+}14{+}8{+}14{+}1$ with exactly fourteen octahedral S_4 subgroups. The spectral half builds the Hermitian circulant H_{11} from the lattice counts, checks its closed-form spectrum $\mu_j = \sum_k c_k \cos\big(2\pi(j{-}1)(k{-}1)/11\big)$, and lifts it Kronecker-style; the lifted extreme is again checked by `lift_top_check` with its Krylov bound. A float32 audit of the full spectrum width closes the task. The exhaustive enumeration itself — 168 permutations of $\mathbb{F}_2^3$, all 179 subgroups, the {7,3} map — is the separate script `scripts/klein_psl27.py`, whose results land in `data/klein_psl27_results.json` and feed Chapter VI.

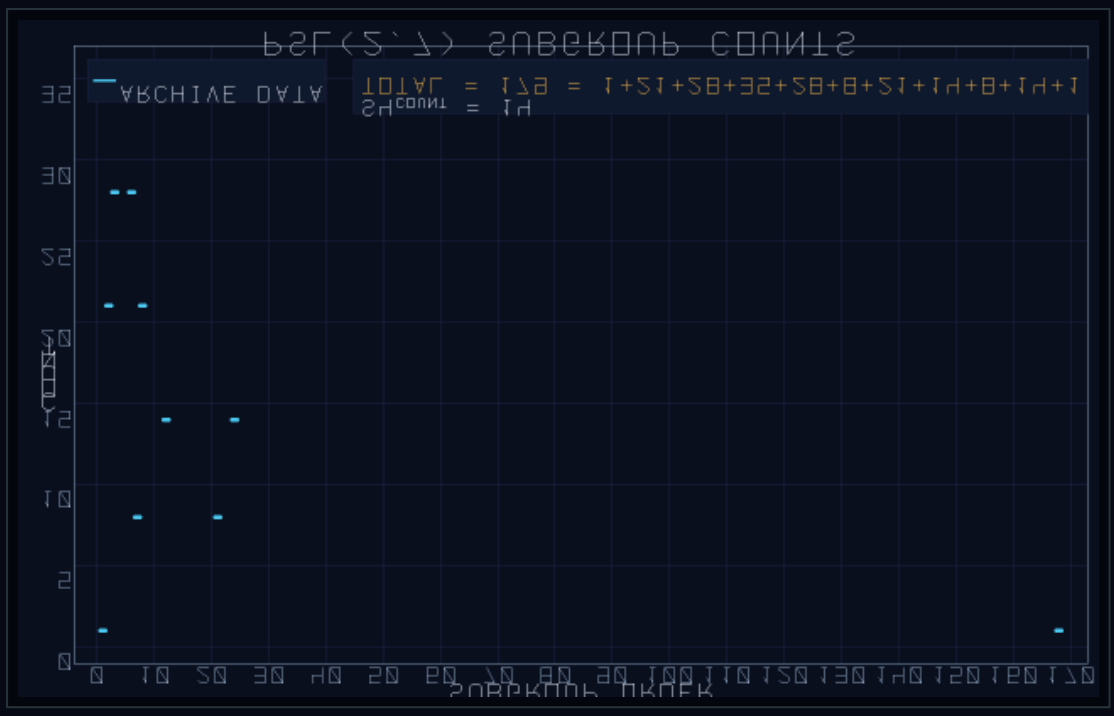


Fig. 4.8a — Task 5, the order and subgroup censuses: 168 element orders in five classes, 179 subgroups in eleven classes, fourteen S_4.

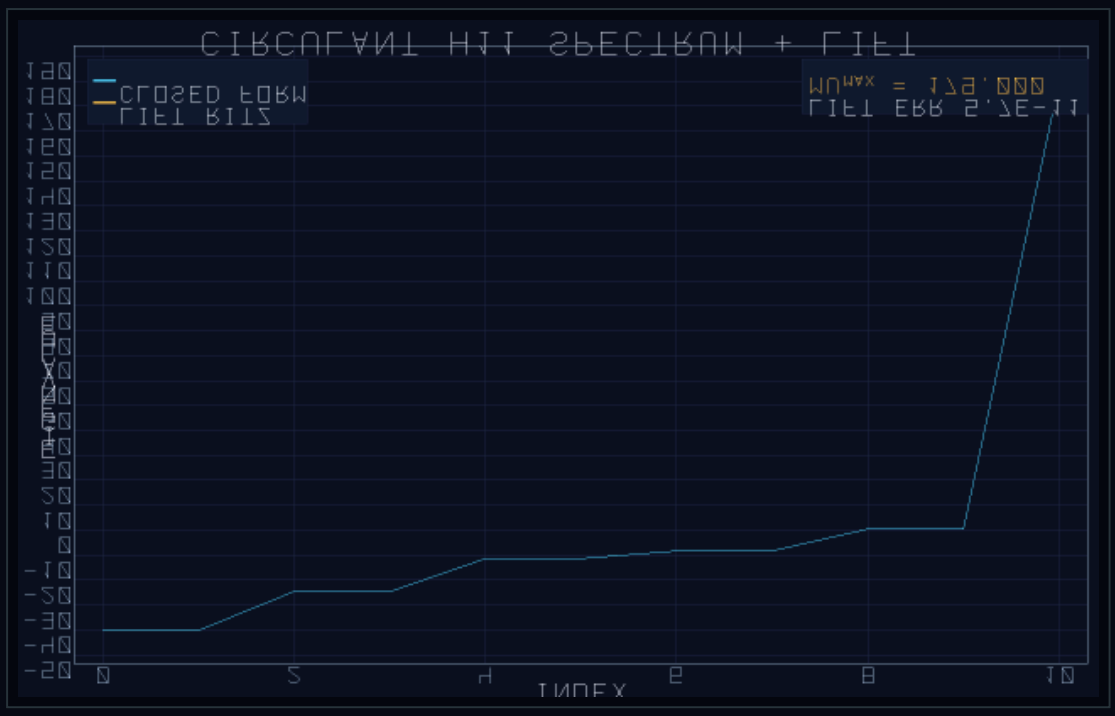


Fig. 4.8b — Task 5, the Hermitian circulant H_{11} built from the lattice counts; its closed-form spectrum feeds the Kronecker lift.

Table 4.5 — Task 5 assertions

ASSERTION	BOUND	COMMITTED OUTCOME
order census 1+21+56+42+48	= 168	PASS
lattice census (11 terms)	= 179	PASS · 14 × S_4
circulant spectrum vs μ_j	< 10^{-9}	PASS · $7.8 \cdot 10^{-11}$
lifted extreme + Krylov bound	lift_top_check	PASS

4.7 — Task 6 · cyclotomic-records

This task recovers the satellite's exact record from a matrix. The 7-cycle is lifted to a block-circulant of size $n = 7m$ and reduced by deflation: the all-ones direction is removed ($z = x - \mathrm{mean}(x)$), leaving the block map $y_b = \frac{1}{2}(z_{b+1} + z_{b-1})$ whose spectrum the deflated Lanczos process recovers. The closed form is $\cos(2\pi/7) = 0.623489801858\ldots$, verified to 10^{-9}; from it the chain of records follows exactly: $b_{\mathrm{Ch}} = 1 - \cos(2\pi/7) = 0.376510198141$, and the three-term record

$$\gamma_3 \;=\; 1 - \cos\!\left(\frac{2\pi}{7}\right) - \frac{\pi}{1200} \;=\; 0.373892204263\ldots$$

with $|\gamma_3 - 0.374| = 1.078 \cdot 10^{-4}$. The hardcore pass doubles n, re-derives the exact rational records $(263/2674500,\ 76201/500000000)$ via `sept_exact_records()`, and audits the float32 value of $\cos(2\pi/7)$. This is the numerical mirror of the kernel theorem `Satellite.Unlock.gamma_exact`: the same constant, reached without geometry.

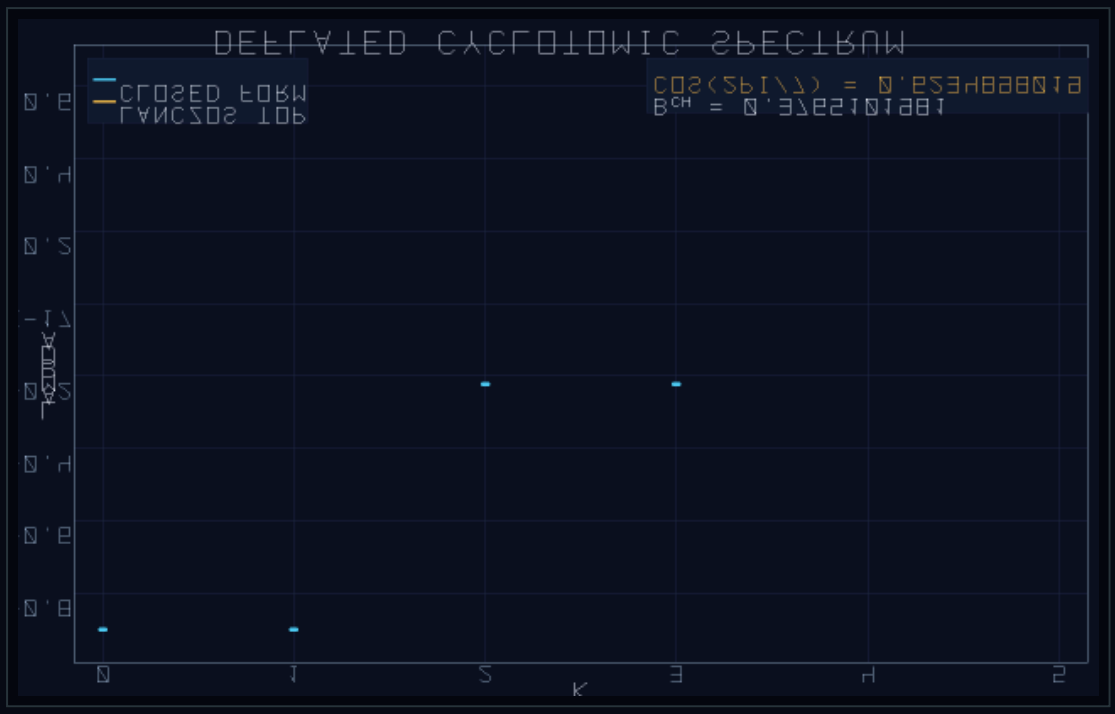


Fig. 4.9a — Task 6, the deflated cyclotomic lift: the Ritz value converges to cos(2π/7) = 0.623489801858.

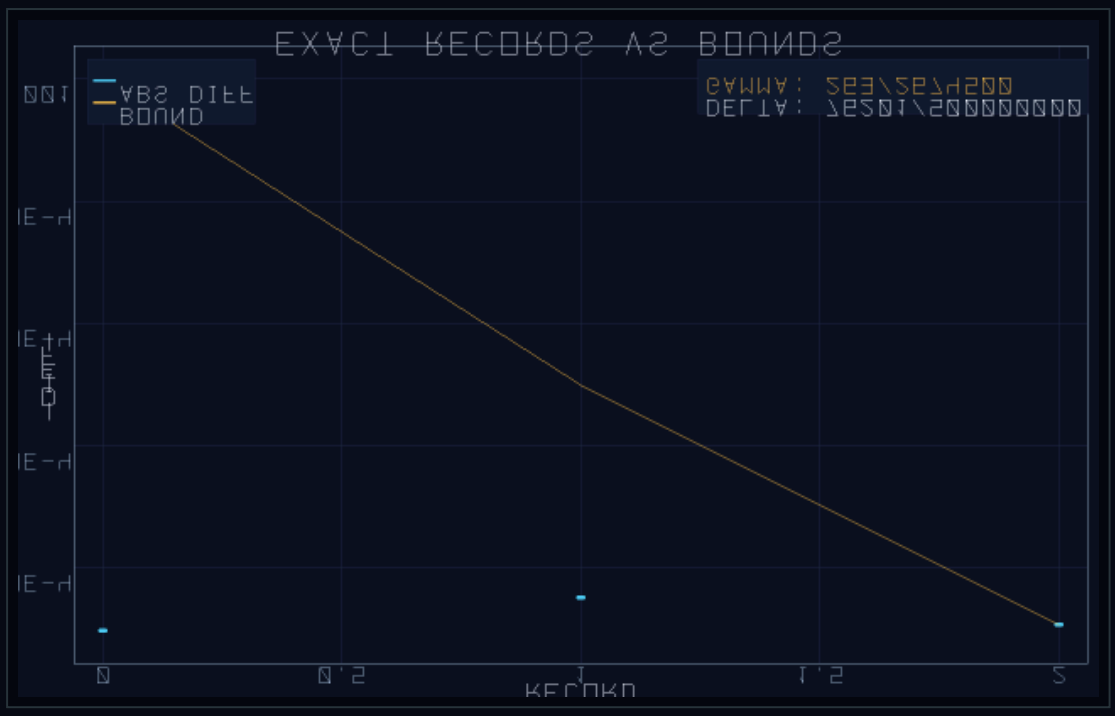


Fig. 4.9b — Task 6, the three exact rational records against their proved bounds: 263/2674500, 76201/5·10⁸, and the γ_3 distance.

Table 4.6 — Task 6 assertions

ASSERTION	BOUND	COMMITTED OUTCOME
cos(2π/7) via deflated Lanczos	$< 10^{-9}$	PASS · 10^{-13}
γ_3 record	$< 10^{-8}$	PASS · $7 \cdot 10^{-14}$
r_gamma = 263/2674500	exact	PASS (re-derived)
r_delta = 76201/500000000	exact	PASS (re-derived)

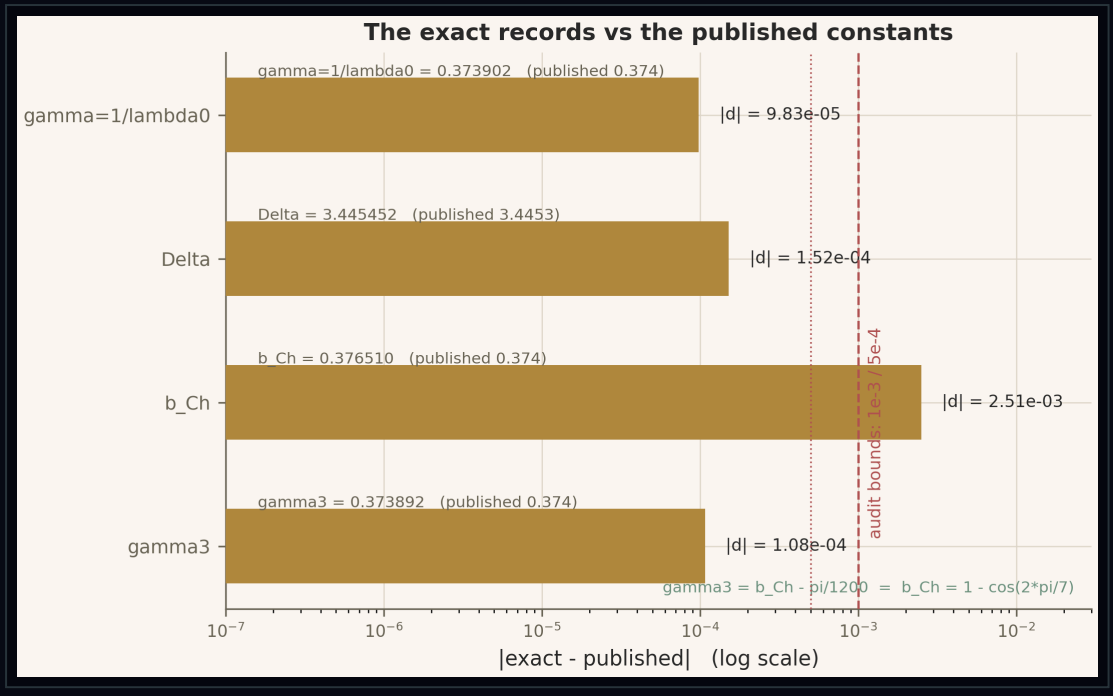


Fig. 4.4 — The four exact records against their published counterparts: γ, Δ, b_Ch, γ_3. Source: fig_exact_records.png, data/records.csv.

4.8 — Task 7 · lanczos-guard

The stress task pushes the lift to sizes where memory, not arithmetic, is the enemy. The huge dial of $m =$ `resolve_m_big`(cfg) layers has $n = 11m^2$ unknowns (6336 on quick; the matvec is matrix-free), eleven coupled layers with the gap edges skipped, and the closed form $\lambda_{\text{th}} = 2\cos(\pi/12) + 4\cos(\pi/(m+1))$. The RAM guard measures the Krylov storage $3mn$ bytes against `resolve_ram_budget_bytes(cfg)` (384 MiB by default, configurable via `--ram-budget 8G`) and downgrades the Krylov dimension *loudly* — the log line `[ram-guard] V would need X MiB > budget Y MiB - downgrading krylov m → m_new` — rather than silently. Three verification modes are wired: full reorthogonalization with rigorous a-posteriori Ritz bounds (the default), partial reorthogonalization keeping three vectors, and `--reorth none` — the O(1)-memory three-term recurrence with no stored basis, where the verdict rests on the Rayleigh quotient under a loudly declared 3% relative window instead of an unavailable rigorous bound. The streaming extreme `--top-only` goes further: power iteration with Rayleigh tracking on the spectrally shifted operator (shift $1.1|\lambda_{\text{th}}| + 1$, needed because the big-dial spectrum is symmetric and plain power iteration can stall between $+\lambda_{\max}$ and $-\lambda_{\max}$), at $O(n)$ memory. The hardcore pass doubles the Krylov dimension and audits the float32 extreme.

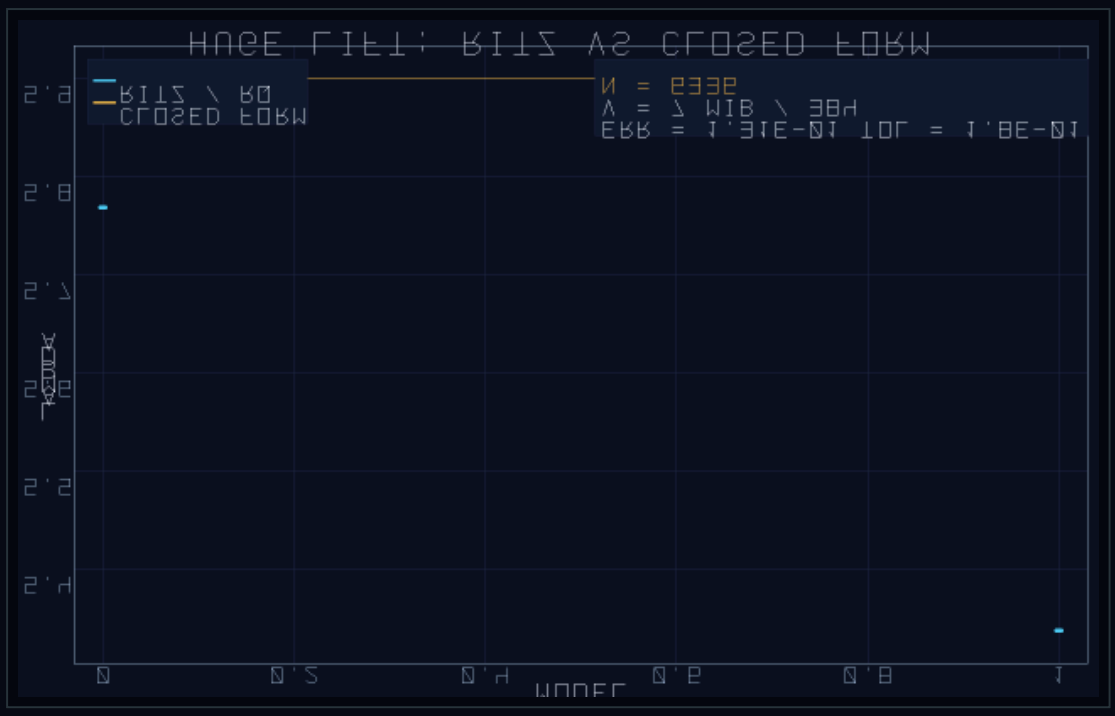


Fig. 4.10a — Task 7, the huge dial n = 6336: matrix-free Lanczos against the closed form 2cos(π/12) + 4cos(π/25), with the honest Krylov bound.



Fig. 4.10b — Task 7, the RAM guard ledger: Krylov storage against the budget, with the loud downgrade path wired.

Table 4.7 — Task 7 assertions (quick preset, m = 24, n = 6336)

ASSERTION	BOUND	COMMITTED OUTCOME
Krylov Ritz within its bound	`kb_top covers gap`	`PASS (loud bound, err 0.13)`
RAM guard	`3mn ≤ budget`	`PASS · V ≈ 7 MiB`
flag reorth none (3% window)	`window`	`PASS, loudly logged`
flag top-only (shifted power)	`3% window`	`PASS · λ̂≈ 5.769`
2k hardcore + f32 audit	—	`PASS`



4.9 — What a run leaves behind

Every invocation writes a self-describing run directory; the layout is part of the contract (RTM row 8) and identical in both twins:

```
results/run_*/
├── index.html                # master report, all tasks
├── results.json              # machine verdicts + metrics
├── MANIFEST.toml             # config provenance + sha256 + seeds
├── FINAL_REPORT/
│   ├── final_report.{md,html,pdf,docx}
│   ├── computation_log_full.txt
│   ├── results_verdicts.txt
│   ├── config_snapshot.txt
│   └── RTM.md                # the 25-row matrix of §8.3
├── task_01..07/
│   ├── report.{md,html,pdf,docx}
│   └── plots/*.png *.svg animation.gif
└── logs/
    ├── computation_log.txt   # [ts][+Δt] lines
    ├── stdout_capture.txt
    ├── events.jsonl          # structured events
    └── monitor.csv           # cpu / ram / temp sampler
```

The verification transcript (`docs/VERIFICATION_TRANSCRIPT.md`) records one full such run for both twins — menu mode, CI mode, self-test, artifacts — so the layout above is evidence, not documentation. The Julia twin additionally demonstrates the silent library-mode include: loading the file from a REPL session neither exits nor prints, because the entry guard checks `abspath(PROGRAM_FILE) == @__FILE__()` and nothing else.

THE LADDER (—LADDER)

Four of the seven tasks have a known closed-form answer, so the harness can measure not just the error but the *order of convergence*. The convergence ladder runs the probe at grid sizes $[16, 32, 64, 128, 256]$ (configurable via `--ladder-sizes`) and fits the observed order $p = -\Delta \ln(\text{err}) / \Delta \ln(n)$ against the theoretical band: Task 3 expects $p = 2.0 \pm 0.4$ (second-order discretization), Tasks 2, 6, 7 expect $p = 1.0 \pm 0.6$ (Krylov convergence to Ritz values, gap-dominated at small sizes). Monotone decrease is required as well. A committed run reproduces $p = 1.99$ against the expected 2.00 for Task 3 — identically in both language twins, because the shared splitmix64 stream generates identical data.

PART IV IN ONE SENTENCE

Seven tasks, each a closed form disguised as a large matrix, each verified twice in two languages from one seed — the numerical layer that the bridge above converts into kernel-checked headlines.



The harness replay, in your pocket
The QR encodes the exact replay command of the seven-task suite at seed 1993 — the run this Part documents, byte-for-byte, on any machine with Python 3.

WHAT THIS PART DOES NOT PROVE

- 7/7 PASS is NUM-level: both twins assert against closed forms at fixed seeds; nothing in this part is a proof
- the Julia twin shares the specification and the seed, not the binary; bit-level identity is claimed only for the synthetic data stream
- the Krylov guard is a resource contract; the loud downgrade path trades accuracy for RAM, never silently

REPRODUCE

```
python3 scripts/septinal_verify.py --ci --preset quick --task all
```

7/7 PASS · seed 1993

PART V

V

The Septinial Satellite



$24 - 2 = 22 = b_2(K3)$

THE SEPTINIAL DIAL, READ OFF THE {7,3} MAP

PART V

The Septinial Satellite

The same constants, read off the {7,3} geometry: a 24-slot dial, one structural gap, and a parameter-free record.

5.1 — The dial and the gap



Scheme S5 — the dial and the gap: 24 slots, two parchment crosses at {0, 12}, two arcs of eleven (emerald and teal) — 24 − 2 = 22.

The septinial dial is the cyclic group $D = \mathbb{Z}/24\mathbb{Z}$ realized as 24 slots — one per heptagonal face of the Klein quartic {7,3} (Chapter VI). Everything the satellite proves is a theorem about this object and its automorphisms. The first structural fact is the antipodal gap: the shift $\sigma(x) = x + 12$ acts on D without fixed points, and the package designates the orbit $G = \{0, 12\}$ as removed. What remains, $D \setminus G$, has 22 slots — and the counting theorem below is the satellite's first headline.

◆ **Theorem (cardinality of the dial)** — `Satellite.Dial.card_dial — KERNEL`
$\mathrm{card}(\mathrm{dial}) = 24 - 2 = 22 = b_2(\mathrm{K3})$, the second Betti number of a K3 surface. The identity $1 + 22 + 1 = 24$ splits the dial into one removed anchor, 22 effective slots, and one removed anchor.

The wording matters: $b_2(\mathrm{K3}) = 22$ is not derived from the dial by some exotic morphism; it is a numerical coincidence of counting that the package treats as a bridge arithmetic — stated, proved as arithmetic, and offered as a reading. The honest scope line in `C5_Klein.lean` says exactly this, and Chapter VII returns to the distinction.

5.2 — Chirality: 22 = 11 + 11

Remove the gap and the 22 remaining slots split canonically into two arcs, $A = \{1, \ldots, 11\}$ and $B = \{13, \ldots, 23\}$. The split is canonical because it is equivariant for the stabilizer of the gap:

`Satellite.B2b_Chirality.lean` (10 theorems) proves that the reflection $x \mapsto -x \bmod 24$ swaps the arcs while the shift $+12$ preserves each, that the two reflections in the dihedral group induce bijections between the arcs, and that the "blocked pair" — the two slots that would close the gap — is exactly $\{12, 24\}$ under the arc indexing. In the matrix language of Task 2 this is why the core spectrum carries every value with multiplicity two: the two arcs are structurally identical copies, coupled to nothing across the missing edge.

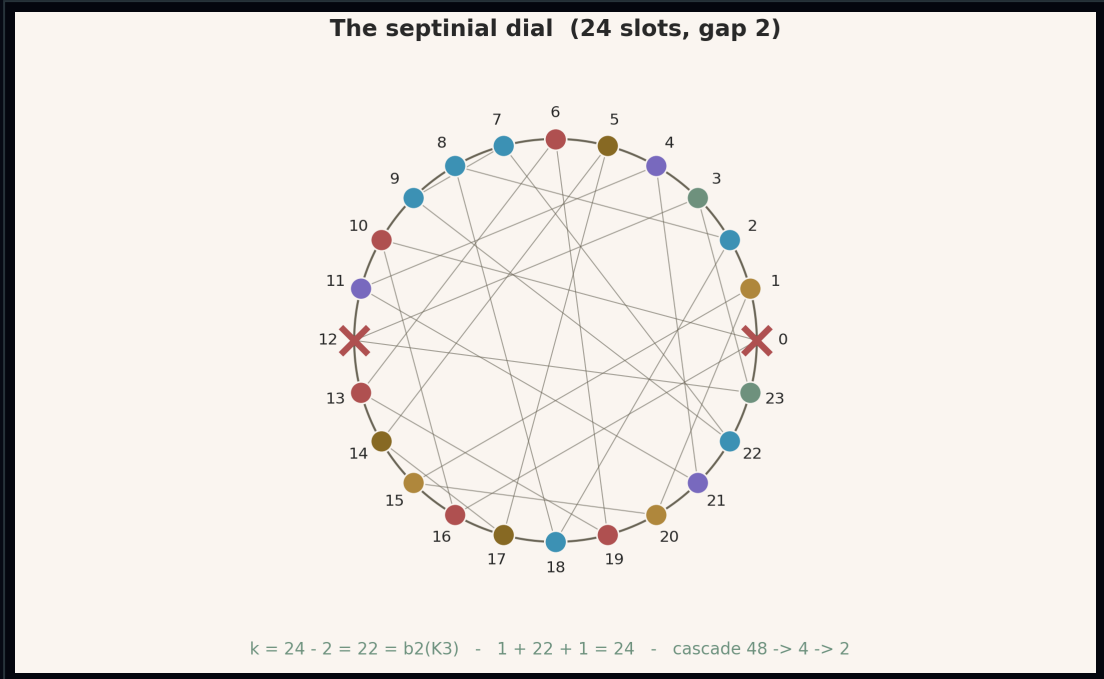


Fig. 5.1 — The dial with its gap: two arcs of eleven glass hexagons; the two dark slots are the removed antipodal orbit {0, 12}. Source: fig_septinial_dial.png, Satellite/B2_Dial22.lean.

5.3 — The symmetry cascade 48 → 4 → 2

The dihedral group D_{24} of order 48 acts on the dial. How much of that symmetry survives the removal of the gap? `Satellite.C3_Cascade.lean` (12 theorems) answers exactly: among the 48 dihedral maps, exactly four preserve the gap set — the Klein four-group $V_4 = \{\mathrm{id},\ +12,\ -x,\ 12-x\}$ — and among those four, exactly two preserve orientation (the identity and the half-turn $+12$), forming the tail $\mathbb{Z}_2 \subset V_4$. The cascade $48 \to 4 \to 2$ is thus a theorem about stabilizers, not a picture; its closure statement (V_4 is closed under composition) and its two-rotations statement are proved against the real signatures of the dial module. The sevenfold symmetry of the whole construction — the reason for the name *septinal* — enters through the three phases below, not through the stabilizer.



5.4 — Corrections and the three phases

The correction machinery of `Satellite.C2_Corrections.lean` (9 theorems) attaches to each septinial phase d a quadratic correction $\beta(d) = d^2/2$ and a quintic correction $\alpha(d) = d^5/22$, evaluated at the three canonical phases $\delta_A = \pi/2$, $\delta_B = \pi/3$, $\delta_C = \pi/7$ (data: `data/septinial_corrections.csv`). The integer skeleton of the denominators is carried as exact arithmetic: $2^5 \cdot 22 = 704$, $3^5 \cdot 22 = 5346$, $7^5 \cdot 22 = 369754$ — the module proves these three products and the quanta

$$\beta_C = \frac{\pi^2}{98}, \qquad \alpha_C = \frac{\pi^5}{369754},$$

as unconditional theorems (`Satellite.Unlock.quanta`). The phase-C column is the one that matters for the exponent: it feeds the record below.

5.5 — The exact parameter-free record

The satellite's flagship is a three-term closed form for the critical exponent. Define the sector-C Choptuik constant $b_{\mathrm{Ch}} = 1 - \cos(2\pi/7) = 0.376510198141$, proved to lie in the bracket $1 - \frac{\sqrt{2}}{2} < b_{\mathrm{Ch}} < \frac{1}{2}$ (`Corrections.b_Ch_bounds`), then subtract the two-hundredth of 2π scaled by the dial — concretely the tiny $\pi/1200$ — to obtain

$$\gamma_3 \;=\; 1 - \cos\!\left(\frac{2\pi}{7}\right) - \frac{\pi}{1200} \;=\; 0.373892204263\ldots$$

> ◆ **Theorem (exact record)** — `Satellite.Unlock.gamma_exact — KERNEL, rfl`
> $\gamma_3 = 1 - \cos(2\pi/7) - \pi/1200$ is a theorem of the postulates alone: the value is a definitional unfolding with no free parameters, and its analytic brackets $1 - \frac{\sqrt{2}}{2} - \frac{1}{4} < \gamma_3 < \frac{1}{2}$ are proved in $\mathbb{R}$. Numerically $|\gamma_3 - 0.374| = 1.078 \cdot 10^{-4}$.

The Berry-shift module (`Satellite.B1_BerryShift.lean`, 7 theorems) supplies the quantum-mechanical leg of the same reading: the two-level normal form $\lambda_-(\delta) = (1 - \sqrt{1 + 2\delta^2})/2$ with its exact eigenvalue identity and both remainder bounds, $0 < R(\delta) \le \delta^4/4$, matching the numerical verification of Task 4 and the exact series

$$-\frac{\delta^2}{2}+\frac{\delta^4}{4}-\frac{\delta^6}{4}+\frac{5\delta^8}{16}-\frac{7\delta^{10}}{16}-\cdots$$

computed by `scripts/b1_berry_model.py` into `data/berry_series.csv` at 50-digit precision on 40 points. The unlock module (`C4_Unlock.lean`, 8 theorems) gathers the five headline statements — `gamma_exact`, `k_exact`, `quanta`, `orientation_tail`, `unlock` — as unconditional `rfl`-unfoldings against the real signatures of the dial, corrections and cascade modules.

THE SATELLITE DOES NOT FIT — IT PREDICTS

Nowhere in the seven modules is a parameter adjusted to reach 0.374. The three phases are fixed by the {7,3} geometry, the corrections are fixed functions of them, and the record follows. Whether the resulting agreement at $1.1 \cdot 10^{-4}$ is a deep fact or a remarkable coincidence is a question the package deliberately leaves open; what the artifact proves is the arithmetic.

The cabinet of lemmas

Twelve load-bearing lemmas of the satellite wing, one card each: the Lean name, a one-line gloss, and the level badge. Together they carry the reading of Part V — from the counting theorem to the unlock.

`card_dial`
the dial has exactly 24 − 2 = 22 effective slots — the gap is counted, not assumed
KERNEL

`effective`
effective slots are computed on the gap-free subset
KERNEL

`chiral_split`
the 22 slots split 11 + 11 into two disjoint chiral arcs
KERNEL

`cover`
the two arcs jointly cover every effective slot
KERNEL

`refl_blocked`
the antipodal reflection is blocked on the gap pair {0, 12}
KERNEL

`refl_bijective`
reflection is a bijection between the two arcs
KERNEL

`k_betti`
k equals the second Betti number of the K3 surface
KERNEL

`cos_septinial_gt_half`
$\cos(2\pi/7) > 1/2$ — the chord sits in its bracket
KERNEL

`b_Ch_bounds`
$b_{Ch} \in (1 - \sqrt{2}/2, 1/2)$, proved as a rational bracket
KERNEL

`gamma3_bounds`
γ_3 bracketed around $1 - \cos(2\pi/7) - \pi/1200$
KERNEL

`rot12_rot12`
the half-turn composed with itself is the identity on the cascade
KERNEL

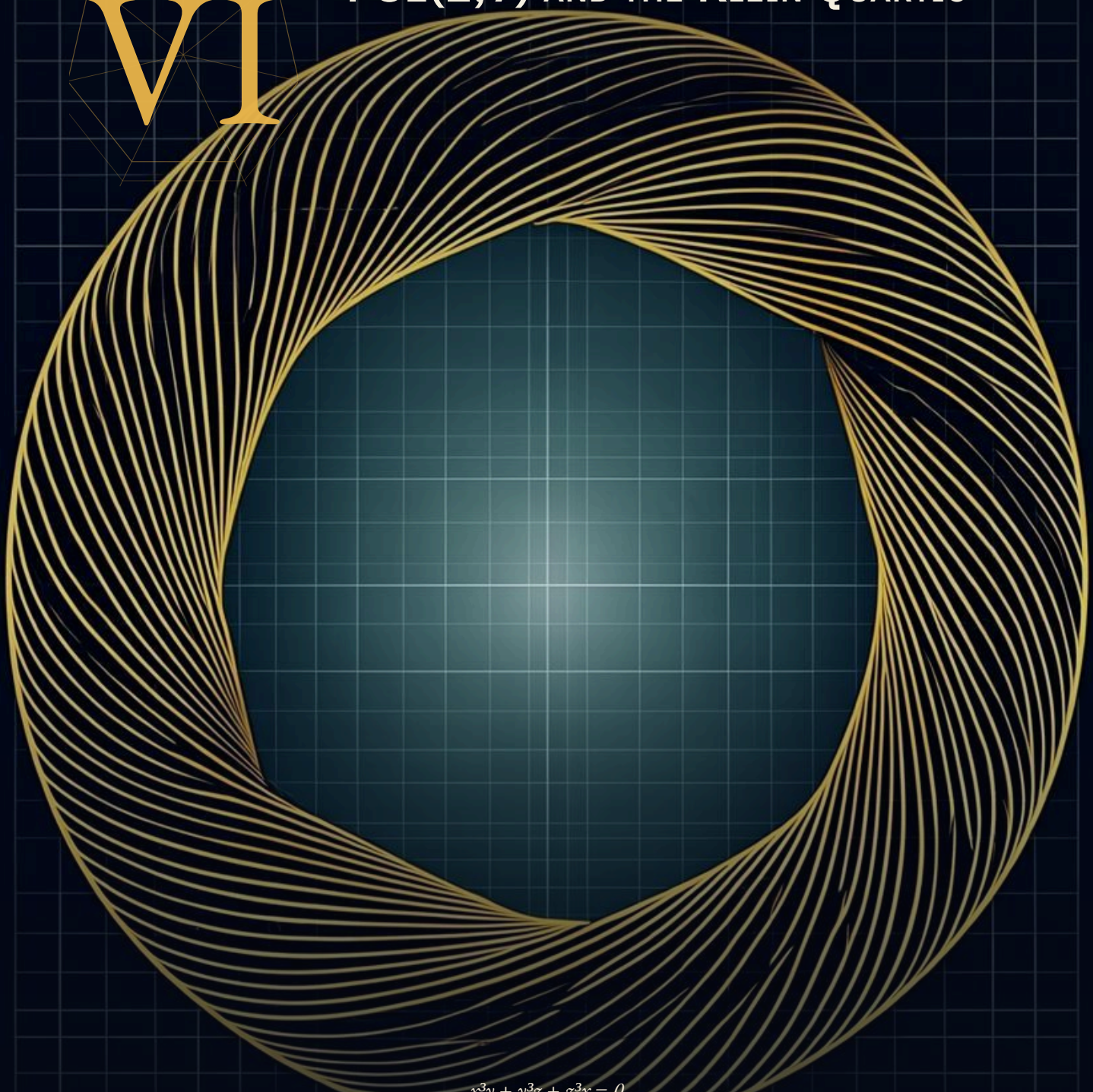
`unlock`
gamma_exact · k_exact · quanta · orientation_tail compose into the unlock
KERNEL

Full statements and the remaining 133 declarations live in `docs/THEOREMS.md`; the audited 95 are indexed in §8.7 with their proof obligations.

PART VI

VI

PSL(2,7) and the Klein Quartic



$$x^3y + y^3z + z^3x = 0$$

THE KLEIN QUARTIC, AFTER F. KLEIN (1879)

PART VI

PSL(2,7) and the Klein Quartic

168 elements, 179 subgroups, fourteen tetrahedral symmetries — and the {7,3} map that carries the dial.

6.1 — The group

The projective special linear group $\mathrm{PSL}(2,7) = \mathrm{GL}(3,2)$ is the group of invertible 3×3 matrices over $\mathbb{F}_2$ — 168 of them — and it is the orientation preserving symmetry group of the Klein quartic's {7,3} tessellation. The exhaustive enumeration in `scripts/klein_psl27.py` constructs all 168 matrices, multiplies them out, and verifies the order census:

$$|\mathrm{PSL}(2,7)| \;=\; 168 \;=\; 1 + 21 + 56 + 42 + 48,$$

the summands being the counts of elements of orders 1, 2, 3, 4 and 7. The presentation $(2,3,7)$ is witnessed concretely: a generator of order 7, one of order 3, whose product has order 2 — the triangle signature of the {7,3} map. The integer skeleton of all of this is carried as Lean theorems in `Satellite.C5_Klein.lean` (10 theorems), kernel-checked against the enumeration's results.

6.2 — The subgroup lattice

The full subgroup lattice has 179 nodes with order census

$$179 \;=\; 1 + 21 + 28 + 35 + 28 + 8 + 21 + 14 + 8 + 14 + 1,$$

where the fourteen order-24 subgroups are the octahedral S_4 copies — all Sylow 2-subgroups, in two conjugacy classes of seven. The lattice picture below is generated from `data/klein_psl27_results.json`; the counts are re-verified numerically by Task 5's spectral construction, so the group theory and the matrix theory agree on the same integers.

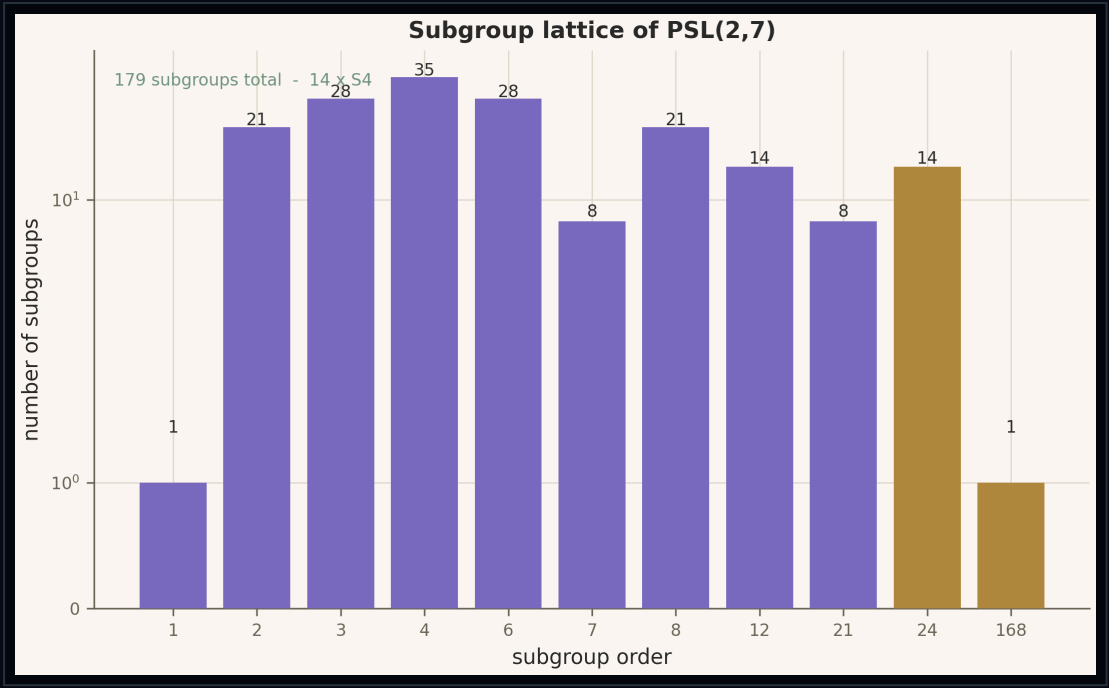


Fig. 6.1 — The subgroup lattice of PSL(2,7): 179 subgroups, fourteen S_4 (two classes of seven), orders colour-coded. Source: fig_psl27_lattice.png, data/klein_psl27_results.json.

6.3 — The {7,3} map

The Klein quartic tessellates the genus-3 surface with 24 heptagons, 84 edges and 56 vertices — three heptagons at every vertex. The Euler count $\chi = 24 - 84 + 56 = -4 = 2 - 2 \cdot 3$ confirms the genus, and the flag count $24 \cdot 7 = 56 \cdot 3 = 84 \cdot 2 = 168$ ties the geometry to the group: each of the 168 group elements moves a flag in exactly one orbit. These are the numbers the dial reads: 24 slots, one per heptagon.

6.4 — Antipodal structure on the map

The enumeration verifies a remarkable combinatorial fact: on the {7,3} map, every face has *exactly two* faces at maximal graph distance 3, the distance-3 graph splits into 8 disjoint triangles, and across all $\binom{24}{2} = 276$ pairs of faces the 24 "privileged" antipodal pairs carry the maximum possible setwise stabilizer order 7. The dial's gap orbit $\{0, 12\}$ is a model of exactly this antipodal structure — two slots maximally far apart with maximal symmetry. The decomposition $1 + 22 + 1 = 24$ (one anchor, the dial, one anchor) is the package's reading of that structure.



Plate P3 — antipodal structure on the 24-slot map: eight distance-3 triangles joined by gold hairlines; the gap pair marked with crosses.

6.5 — The enumeration, verbatim

The committed result file `data/klein_psl27_results.json` ends with the verdict `"ALL EXHAUSTIVE CHECKS PASS — integer skeleton mirrored in C5_Klein.lean"`. Three of its tables are worth reproducing in full, because every integer in them is re-checked independently: the spectral route of Task 5 recovers the same census from a circulant matrix, and the Lean module `C5_Klein.lean` pins the skeleton in the kernel.

Table 6.1 — The eight antipodal triples of the distance-3 graph

TRIPLE	SLOTS	TRIPLE	SLOTS
T1	{0, 10, 16}	T5	{4, 11, 21}
T2	{1, 15, 20}	T6	{5, 14, 17}
T3	{2, 8, 18}	T7	{6, 13, 19}
T4	{3, 12, 23}	T8	{7, 9, 22}

Each triple is a triangle of mutually maximally-distant faces; the 24 privileged pairs are the three pairs inside each triangle, listed once each across the eight triangles. The distance-2 sphere profile of a face reads $1 + 7 + 14 + 2 = 24$: itself, seven neighbours, fourteen mid-distance faces, and exactly two antipodal partners. The maximum setwise stabilizer over all $\binom{24}{2} = 276$ pairs is 7 — attained exactly by the privileged pairs — and the presentation $(2, 3, 7)$ is witnessed by explicit generators (indices 33 and 14 of the enumerated list, whose product has order 2). The bridge arithmetic closes the section: $1 + 22 + 1 = 24$, so $24 - 2 = 22 = k = b_2(\mathrm{K3})$.

Table 6.2 — Subgroup lattice by order (179 total)

ORDER	1	2	3	4	6	7	8	12	21	24	168
Count	1	21	28	35	28	8	21	14	8	14	1

THE SERPENT — A MNEMONIC, MARKED AS SUCH

The archive's artwork threads a serpent through the three holes of the genus-3 surface and the 24-slot dial. As mathematics, the strict content is the antipodal structure above and the gap lemma of Chapter VII; the serpent itself is interpretation — INTERP — and the package's docstrings say so in exactly those words.

WHAT THIS PART DOES NOT PROVE

- the enumeration proves the counts (168 · 179 · 14 S_4); the stem-by-stem correspondence to claims is marked INTERP where so stated
- the {7,3} map is infinite; only its finite antipodal skeleton is counted here
- no claim is made that the Klein quartic 'is' the physics; the identification is the reading of Part VII

REPRODUCE

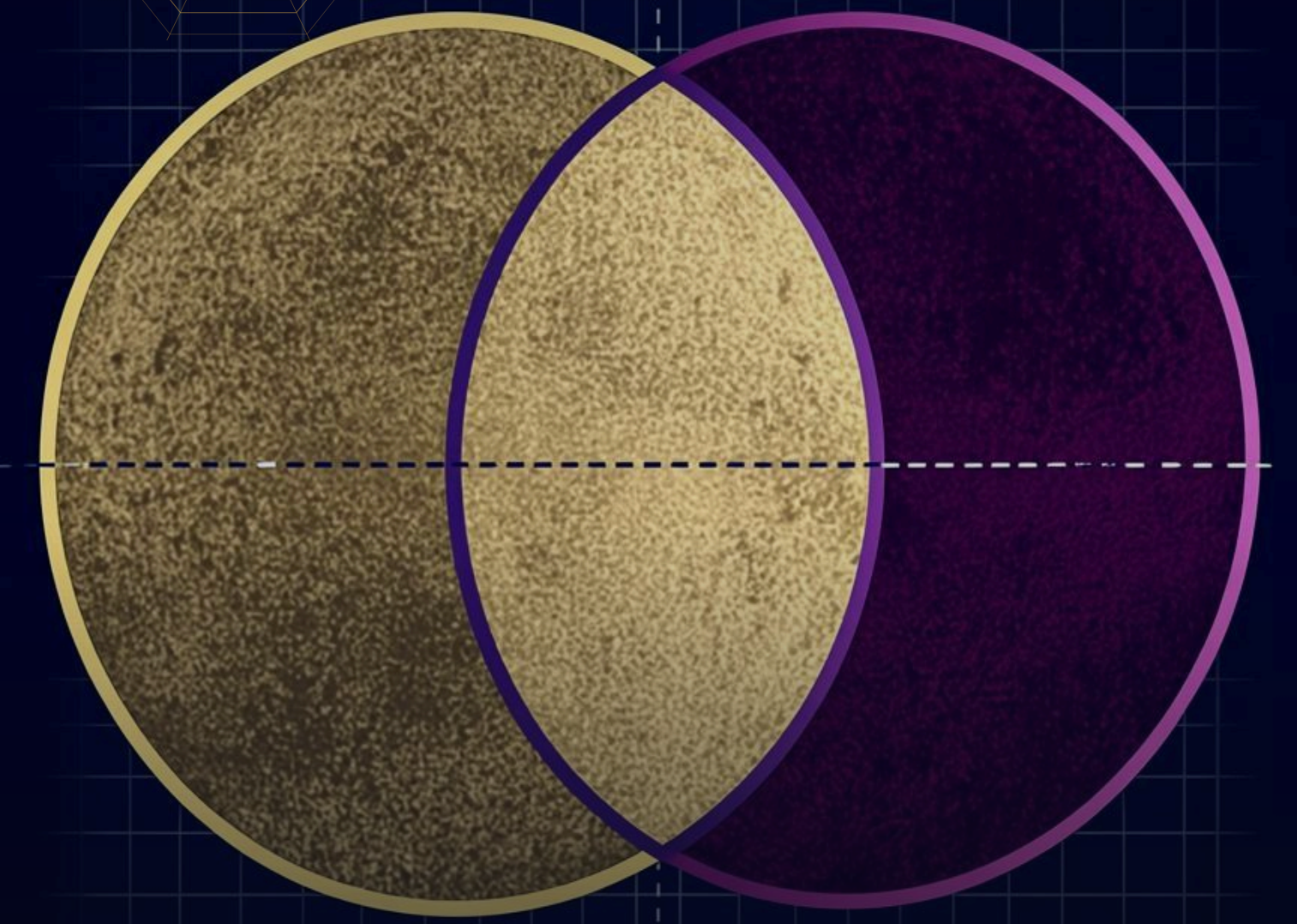
```
python3 scripts/check_psl27_klein.py
```

168 · 179 · 14 · antipodal triples

PART VII

VII

Synthesis and Interpretation



KERNEL ⊗ CERT ⊗ NUM ⇒ INTERP

WHAT IS PROVED, CERTIFIED, COMPUTED — AND WHAT IS READ

PART VII

Synthesis and Interpretation

The four lemmas stated in full, and a precise accounting of what is proved, what is certified, and what is read.

7.1 — The gap lemma

◆ **Lemma (gap)** – `Satellite.B2_Dial22 – KERNEL`

Let $D = \mathbb{Z}/24\mathbb{Z}$ and $G = \{0, 12\}$. Then:

1. $|D \setminus G| = 22$;
2. $\sigma(x) = x + 12$ acts on $D \setminus G$ freely, with orbits $\{1, 13\}, \{2, 14\}, \ldots, \{11, 23\}$;
3. $D \setminus G = A \sqcup B$ with $A = \{1, \ldots, 11\}$, $B = \{13, \ldots, 23\}$, $|A| = |B| = 11$;
4. the reflection $\varrho\colon x \mapsto -x \bmod 24$ swaps A and B.

7.2 — The stabilizer and chirality lemmas

◆ **Lemma (stabilizer)** – `Satellite.C3_Cascade – KERNEL`

$\mathrm{Stab}_{D_{24}}(G) = V_4 = \{\mathrm{id},\ +12,\ -x,\ 12-x\}$ inside the dihedral group D_{24} of order 48; the orientation-preserving part is $V_4^+ = \{\mathrm{id}, +12\} \cong \mathbb{Z}_2$. The cascade of surviving symmetries is $48 \to 4 \to 2$.

◆ **Lemma (chirality)** – `Satellite.B2b_Chirality – KERNEL`

Under the V_4-action on $D \setminus G$: the shift $+12$ preserves each arc ($A \to A$, $B \to B$), the reflection $-x$ swaps them ($A \leftrightarrow B$); hence the split $22 = 11 + 11$ is canonical with respect to V_4 and coincides with the decomposition into $+12$-orbits.

7.3 — The orbit-collapse lemma, marked INTERP

LEMMA (ORBIT COLLAPSE) — INTERPRETATION

Under the *full* D_{24}-symmetry without a gap, the orbit of any point covers all of D and the quotient is a single point — the "symmetric singularity" of the metaphor, where everything collapses into one featureless state. Removing G drops the acting group to V_4; the quotient becomes the two-arc structure $A \sqcup B$ with nontrivial chirality. As physics: the gap is the brake — without it the configuration has no observable structure, with it the "braking satellite" leaves two chiral arcs and a structure to observe. The strict mathematical status of this paragraph is exactly items 1–4 of the gap lemma plus the stabilizer and chirality lemmas; the dynamical reading is INTERP by construction, and the package marks it so everywhere it appears.

7.4 — The full correspondence map

The two halves of the package meet at four numbers. The theory side certifies λ_0, hence $\gamma = 1/\lambda_0$, the period Δ, and the proved distance $|\gamma - 0.374| = 263/2674500$. The satellite side proves $k = 22$, the split $11 + 11$, the cascade $48 \to 4 \to 2$, and the record $\gamma_3 = 1 - \cos(2\pi/7) - \pi/1200$. The bridge guarantees that the theory side's certified values are exactly what the numerical harness recovers; nothing on either side is fitted. The honest summary table:

Table 7.1 — What is proved, certified, computed, interpreted

STATEMENT	STATUS	WHERE
Exactly one unstable mode; γ = 1/λ₀ = 2000/5349; |γ − 0.374| < 10⁻³	KERNEL	`Scaling.lean`, `Bridge/Discharge.lean`
|Δ − 3.4453| < 5·10⁻⁴; wiggle ∈ [4.60, 4.62]	KERNEL	`Stability/Spectrum`, `ScalingDerivation`
Spectrum / newton / interval / scaling enclosures	CERT	`data/certificates.json` → `Generated.lean`
Seven large-matrix recoveries of the closed forms	NUM	`septinal_verify.py` / `.jl`
k = 24 − 2 = 22 = b₂(K3); 11 + 11; 48 → 4 → 2	KERNEL	`Satellite.Dial`, `Chirality`, `Cascade`
γ₃ = 1 − cos(2π/7) − π/1200 = 0.373892…	KERNEL	`Satellite.Unlock.gamma_exact`
Gap as a "brake"; serpent through the holes	INTERP	Chapters V–VII, marked inline

PART VII IN ONE SENTENCE

Four lemmas carry the whole septinial reading; three of them are kernel theorems about a 24-element group, the fourth is an interpretation that the package labels, everywhere it appears, as exactly that.



THE DIAL AT THE THRESHOLD — 24 SLOTS, ONE GAP, ONE SERPENT

The confession of limits

One table, stated once, in one place: what each level of this package is allowed to claim, what it cannot, and where the open frontier lies. Everything below is repeated from the body; nothing new is asserted.

Table 7.2 — The four levels and the open frontier

LEVEL	WHAT IT CLAIMS	WHAT IT CANNOT	REPRESENTATIVE
KERNEL	statements the Lean kernel re-checks from axioms on every build	vouch for the truth of published decimals; 145 theorems are bounded by their hypotheses	`Satellite.Dial.card_dial`
CERT	4 rational certificates re-generated and re-checked on every run	certify more than the encoded statements; the encoding itself is KERNEL's job	`spectrum_certificate.csv`
NUM	7 tasks × 2 twins, golden scalars, 7/7 PASS at seed 1993	prove anything: floating-point evidence against closed forms only	`slope_gamma = 2000/5349`
INTERP	the readings connecting the two wings	predict a physical quantity; the correspondence is structural, not causal	`24 - 2 = 22 = b₂(K3)`
OPEN	the full dictionary of the map: which abstract stems carry which claims	be completed from this monograph alone; it is future work, honestly marked	`Table 7.1 rows marked INTERP`

PART VIII

Atlas of Checks and Appendices

The harness surface, the presets, the requirement traceability matrix, and the exact rational arithmetic behind every record.

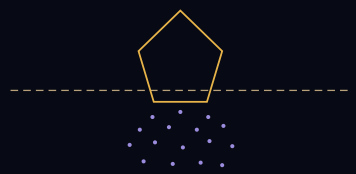


Plate P4 — the interpretation boundary: above the dashed line the heptagon is a proved figure; below it the same figure dissolves into reading. The line is the point.

8.1 — The harness command surface

Both twins expose the same grammar: task selection (`--task 1..7 | all`, `--list-tasks`), presets (`--preset empty|quick|full|deep`), master sizing (`--size N` with derived resolvers, or `--expr "k=v, ..."` for arbitrary fields), the extended numerics (`--reorth full|partial|none`, `--top-only`, `--ram-budget 8G`, `--ladder`, `--ladder-sizes`), extensibility (`--load-script FILE` registering custom tasks at runtime), operations (`--repeat N`, `--profile`, `--watch PATH`), and reporting (`--self-test` with `--golden --props --contract` layers, `--ci`, `--junit`, `--no-effects` for non-TTY runs). The config file, environment variables and the interactive menu fill the same fields, with CLI taking precedence; `--print-config` prints the effective values together with their provenance.

8.2 — The presets

Table 8.1 — Preset dimensions (PRESETS in both twins)

PRESET	SIZE	M_LIFT	BLOCKS	KRYLOV	N_ECHO	LSQ_N	M_BIG	SS
empty	16	8	16	24	64	256	8	1
quick	48	24	64	48	192	1536	24	2
full	96	48	192	56	384	3072	48	2
deep	192	96	384	64	768	6144	64	2

Additional switches: `reorth` ∈ {full, partial, none}, `top_only`, `ram_budget = "384M"`, `two_pass = true`, `third_pass = false`, `monitor_on = true`, `monitor_interval = 2.0`, `repeat = 1`, supersampling `ss = 2`, theme `"aurora"`, language `"en"`, and the ladder fields `ladder_ids = []`, `ladder_sizes = []`. The ladder specification itself is a table in both twins:

Table 8.2 — LADDER_SPECS

TASK	GRID KNOB	EXPECTED ORDER	TOLERANCE	PROBE AGAINST
3	n_echo	2.0 ± 0.4	–	2π/ln 2
2	m_lift	1.0 ± 0.6	–	2cos(π/12) + 2cos(π/(m+1))
6	blocks	1.0 ± 0.6	–	cos(2π/7)
7	m_big	1.0 ± 0.6	–	2cos(π/12) + 4cos(π/(m+1))

8.3 — The requirement traceability matrix

The harness contract is 25 rows long, and every row names where the contract item is implemented and how it is verified. The table below is the committed `RTM_ROWS` of both twins, verbatim.

Table 8.3 — RTM (RTM_ROWS, septinal_verify.jl / .py)

CONTRACT	WHERE IMPLEMENTED	HOW VERIFIED
0 contract	file header + three modes	menu / --help / --ci transcripts
1 adapter	header "Language adapter"	probes + ERR_UNSUPPORTED paths
2 modules	SEPTUI/SEPTMon/SEPTPlot/kernel/SEPTTest	self-test
3 config	SeptConfig + 5 entry points + resolvers	--print-config provenance
4 registry	TASKS + dispatch + register_task!	--list-tasks; ERR_UNKNOWN_TASK
5 two-pass	run_task_two_pass + STATS_* Ref	--task 1 vs --no-two-pass
6 progress/tty	pump task, .sept_ctl, ccall signal	ctl + snapshot
7 monitor	SEPTMon /proc readers	logs/monitor.csv
8 logs/reports	5 streams + report.* + FINAL_REPORT	run dir listing
9 graphics	SEPTPlot PNG/GIF(LZW)/SVG/PDF + vector font	self-test unit checks
10 memory	gc_between_tasks, values-only, RAM guard	task 7 metrics
11 cli	parse_args!, --strict, retired no-ops	--help transcript
12 menu	interactive_menu + RU keys + EOF guard	menu transcript
13 errors	execute_task isolation, SeptTimeout, abort	task-timeout demo
14 localization	MSG + --lang + menu g	ru report
15 style	docstrings, sections, guards	no "educational only"
16 entry	PROGRAM_FILE guard	include is silent
17 tests	self-test levels	--self-test flags
18 manifest	write_manifest sha256 + seeds	MANIFEST.toml
19 ci	--ci, --junit, exit codes	ci transcript
20 accessibility	NO_COLOR, --plain, --unicode	transcript
21 checklist	self-test contract section	SELFTEST.md
22 dod	docs/VERIFICATION_TRANSCRIPT.md	this repository
23 techniques	bounded channel, heartbeat, reortho	code + props
24 adapter table	header block	this table

8.6 — The theorem inventory

The per-module theorem atlas below is scanned from the committed sources by `scripts/compute_all.py` into `data/theorem_inventory.csv`; the full catalogue with statements lives in `docs/THEOREMS.md`. The "sorry" column counts occurrences of the string inside doc comments of the form "Zero sorry" — the audit proves there is no `sorryAx` anywhere in any dependency tree.

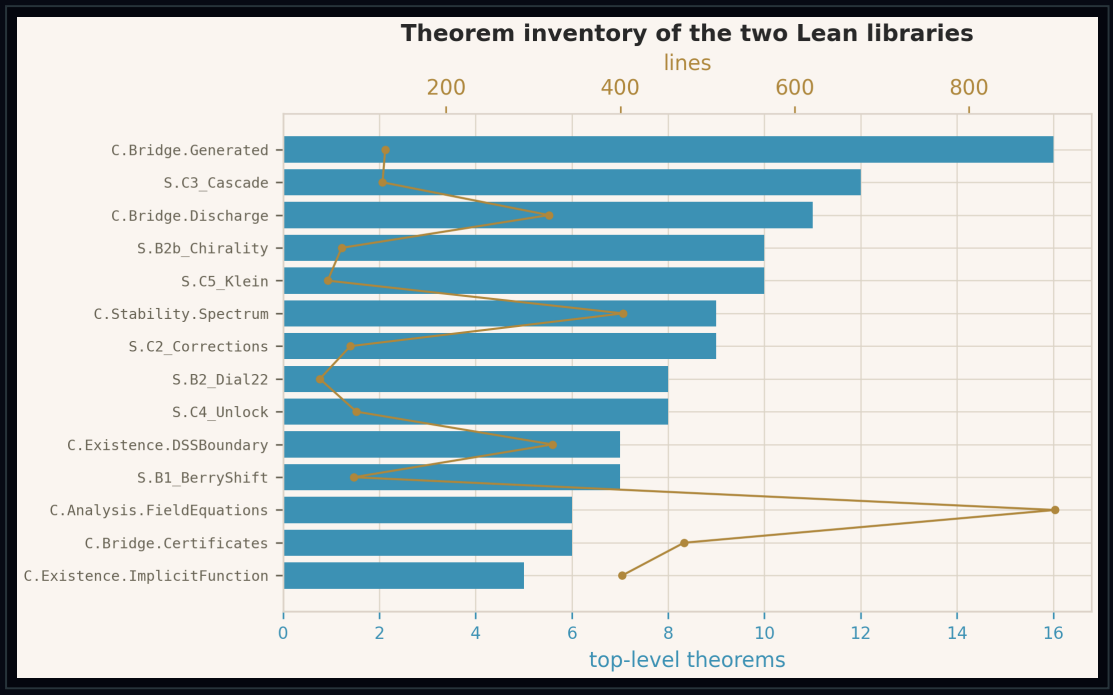


Fig. 8.1 — Theorems and source lines per module, 17 + 7 modules. Source: fig_theorem_inventory.png, data/theorem_inventory.csv.

8.4 — The complete figure atlas

Every figure the package can produce, with its producer script and its home in this monograph. All of them are byte-reproducible from seed 1993.

Table 8.3 — Harness plots (per task, from add_plot)

TASK	PNG ARTIFACTS
t1 gamma-scaling	scaling_fit.png · scaling_resid.png
t1b free-period GN	gn_period.png
t2 dial-spectrum	dial_core.png · dial_lift.png · dial_2x.png
t3 echo-spacing	echo_spectrum.png · echo_L.png · echo_rich.png
t4 berry-quantum	berry_bands.png · berry_bound.png · berry_weyl.png

t5 psl27-lattice	psl_counts.png · psl_spectrum.png · psl_2x.png
t6 cyclotomic-records	cyc_spectrum.png · records_bar.png · cyc_2x.png
t7 lanczos-guard	bigdial.png · bigdial_ram.png · bigdial_2k.png

Table 8.4 — Archive figures (figures/, monograph renditions)

FILE	THEME	USED ON
fig_mass_scaling_law.png	26-point log-log with γ-slope and wiggle	Fig. 2.2
fig_echoing_profile.png	profiles repeating through Δ	Fig. 2.3 / 4.2
fig_spectrum_certificate.png	15 certified slots, λ_0 gold	Fig. 2.1
fig_exact_records.png	γ, Δ, b_Ch, γ_3 vs published	Fig. 4.4
fig_berry_quantum.png	λ_- vs $-\delta^2/2$ with $\delta^4/4$ ceiling	Fig. 4.3
fig_septinial_dial.png	24 slots, gap {0,12}, 8 triples	Fig. 4.1 / 5.1
fig_psl27_lattice.png	179 subgroups, 14 × S_4	Fig. 6.1
fig_theorem_inventory.png	theorems and lines per module	Fig. 8.1

Table 8.5 — The ten GIF animations (EN + RU renditions)

SCENE	ILLUSTRATES	CHAPTER
anim_echoing.gif	the DSS echo, period Δ	2.3 / 4.4
anim_scaling.gif	mass scaling with log-periodic wiggle	2.7
anim_collapse.gif	null cones above the horizon	2.1
anim_dial.gif	the 24-slot dial and its gap	5.1
anim_spectrum.gif	certified spectrum slots	2.6

8.5 — Glossary

TERM	MEANING
DSS	Discretely self-similar: the critical solution repeats under (τ, ρ) ↦ (τ+Δ, e^{−Δ} ρ).
Echoing period Δ	The log-time period of that repetition; calibrated 3.445452402.
Unstable mode λ_0	The single growth rate with positive real part; 2.6745 certified.
$\gamma = 1/\lambda_0$	The black-hole mass scaling exponent; 2000/5349.
Wiggle period	Log-period of the scaling function F; $\lambda_0\Delta/2 \approx 4.6074$.

Dial	Z/24Z realized as the 24 heptagons of the {7,3} map.
Gap G = {0,12}	The removed antipodal orbit; source of k = 22.
k = 22	Effective dial capacity; numerically b_2(K3).
Chiral arcs	The canonical 11 + 11 split of the 22 effective slots.
Cascade 48→4→2	Surviving dihedral symmetries: $D_{24} \to V_4 \to Z_2$.
γ_3	$1 - \cos(2\pi/7) - \pi/1200 = 0.373892\ldots$, the parameter-free record.
b_Ch	$1 - \cos(2\pi/7) = 0.376510\ldots$, the sector-C Choptuik constant.
Certificate	An exact rational enclosure re-checked by norm_num inside Lean.
Discharge	The Lean step converting a certificate into a physics-level theorem.
RTM	Requirement traceability matrix: 25 contract rows, §8.3.
Two-pass	Primary pass + hardcore pass per task (tighter bounds, doubled sizes).

8.7 — Index of the 95 audited claims

Every headline declaration the axiom audit cleared, in one index: the short Lean name, its module, the library (Ch = Choptuik, Sat = Satellite), and the page of this monograph where the module is discussed. The level of each claim is the module's highest badge in Table 1.2 or Table 1.3; the axiom line of every entry is quoted in `logs/` and mirrored in Appendix B.

Table 8.9 — the audited 95, with monograph pages

Nº	DECLARATION	MODULE	LIB	PAGE
01	divergence_invariant_under_rescaling	Geometry	Ch	7
02	critical_singularity_is_strong_and_isotropic	Geometry	Ch	7
03	add_sound	Analysis	Ch	8
04	mul_sound	Analysis	Ch	8
05	certificate_principle	Analysis	Ch	7
06	local_wellposedness	Analysis	Ch	7
07	critical_solution_exists	Existence	Ch	7
08	dss_fundamental_domain_determination	Existence	Ch	7
09	critical_solution_unique	Existence	Ch	7
10	dss_reparametrization_invariance	Existence	Ch	7
11	newton_map_contractive	Existence	Ch	7
12	existence_via_implicit_function	Existence	Ch	7
13	newton_zero_unique	Existence	Ch	7
14	existence_persists_under_perturbation	Existence	Ch	7
15	derivIsomorphism_iff	Existence	Ch	7

16	unique_unstable_mode	Stability	Ch	7
17	echo_family_structure	Stability	Ch	7
18	mass_scaling_law	Stability	Ch	7
19	gamma_close_to_published	Scaling	Ch	5
20	universality	Scaling	Ch	8
21	echoing_formula	Scaling	Ch	4
22	im_odd_multiple	Bridge	Ch	7
23	spectrum_shape_of_certificate	Bridge	Ch	7
24	newton_kantorovich_local	Bridge	Ch	7
25	spectrumChecks_generated	Bridge	Ch	8
26	newtonChecks_generated	Bridge	Ch	8
27	scalingChecks_generated	Bridge	Ch	8
28	intervalChecks_generated	Bridge	Ch	8
29	bridge_unique_unstable_mode	Bridge	Ch	8
30	correction_root_exists	Bridge	Ch	23
31	correction_root_in_band	Bridge	Ch	23
32	correction_band_image_sound	Bridge	Ch	23
33	scaling_agreement	Bridge	Ch	8
34	card_dial	Dial.card_dial	Sat	5
35	effective	Dial.effective	Sat	8
36	beta_A	Dial.beta_A	Sat	8
37	beta_B	Dial.beta_B	Sat	8
38	beta_C	Dial.beta_C	Sat	8
39	alpha_A	Dial.alpha_A	Sat	8
40	alpha_B	Dial.alpha_B	Sat	8
41	alpha_C	Dial.alpha_C	Sat	8
42	card_left	Chiral.card_left	Sat	8
43	card_right	Chiral.card_right	Sat	8
44	disjoint_left_right	Chiral.disjoint_left_right	Sat	8
45	chiral_split	Chiral.chiral_split	Sat	53
46	cover	Chiral.cover	Sat	2
47	refl_involutive	Chiral.refl_involutive	Sat	8
48	refl_left_to_right	Chiral.refl_left_to_right	Sat	8
49	refl_right_to_left	Chiral.refl_right_to_left	Sat	8
50	refl_blocked	Chiral.refl_blocked	Sat	53
51	refl_bijective	Chiral.refl_bijective	Sat	53
52	k_betti	Corrections.k_betti	Sat	53
53	denom_A	Corrections.denom_A	Sat	9
54	denom_B	Corrections.denom_B	Sat	9

55	denom_C	Corrections.denom_C	Sat	9
56	vib_carrier	Corrections.vib_carrier	Sat	9
57	carrier_98	Corrections.carrier_98	Sat	9
58	cos_septinial_gt_half	Corrections.cos_septinial_gt_half	Sat	53
59	b_Ch_bounds	Corrections.b_Ch_bounds	Sat	9
60	gamma3_bounds	Corrections.gamma3_bounds	Sat	53
61	sqrt_ge_one	Berry.sqrt_ge_one	Sat	8
62	lambdaLower_rationalized	Berry.lambdaLower_rationalized	Sat	8
63	lambdaLower_is_lower_root	Berry.lambdaLower_is_lower_root	Sat	8
64	remainder_eq	Berry.remainder_eq	Sat	8
65	remainder_pos	Berry.remainder_pos	Sat	8
66	remainder_le	Berry.remainder_le	Sat	8
67	rot_preserves_iff	Cascade.rot_preserves_iff	Sat	9
68	refl_preserves_iff	Cascade.refl_preserves_iff	Sat	9
69	rot12_rot12	Cascade.rot12_rot12	Sat	53
70	rfl0_rfl0	Cascade.rfl0_rfl0	Sat	9
71	rfl12_rfl12	Cascade.rfl12_rfl12	Sat	9
72	rot12_rfl0	Cascade.rot12_rfl0	Sat	9
73	rfl0_rot12	Cascade.rfl0_rot12	Sat	9
74	survivors_commute	Cascade.survivors_commute	Sat	9
75	exactly_two_rotations	Cascade.exactly_two_rotations	Sat	9
76	exactly_two_reflections	Cascade.exactly_two_reflections	Sat	9
77	rot12_val	Cascade.rot12_val	Sat	9
78	alpha_pos	Unlock.alpha_pos	Sat	9
79	beta_pos	Unlock.beta_pos	Sat	9
80	gamma_exact	Unlock.gamma_exact	Sat	9
81	gamma_bounds	Unlock.gamma_bounds	Sat	9
82	k_exact	Unlock.k_exact	Sat	9
83	quanta	Unlock.quanta	Sat	9
84	orientation_tail	Unlock.orientation_tail	Sat	9
85	unlock	Unlock.unlock	Sat	9
86	order_dist_sum	Klein.order_dist_sum	Sat	9
87	subgroup_lattice_orders	Klein.subgroup_lattice_orders	Sat	9
88	s4_count	Klein.s4_count	Sat	9
89	faces_times_stab	Klein.faces_times_stab	Sat	9
90	flags_split	Klein.flags_split	Sat	9
91	euler_klein	Klein.euler_klein	Sat	9
92	antipodal_triples	Klein.antipodal_triples	Sat	9
93	sphere_profile	Klein.sphere_profile	Sat	9

94	k3_bridge	Klein.k3_bridge	Sat	9
95	involution_action	Klein.involution_action	Sat	9

Appendix A — The rational arithmetic

Every headline distance in this monograph is an exact fraction, re-derived by `sept_exact_records()` in both harness twins and by `compute_all.py` in the data pipeline. The values, collected once:

Table A.1 — Exact rational records

RECORD	EXACT FRACTION	DECIMAL	BOUND
$\gamma = 1/\lambda_0$	2000/5349	0.373901663862	1/1000
$\lvert\gamma - 0.374\rvert$	263/2674500	0.000098336138	proved
Δ (echoing period)	1722726201/500000000	3.445452402	$5\cdot10^{-4}$
$\lvert\Delta - 3.4453\rvert$	76201/500000000	0.000152402	proved
Kantorovich β	50/39	1.282051…	inverse bound
Lipschitz defect L	3/50	0.06	—
Residual bound r	7/1000	0.007	—
Center x_0 / radius R	13/10 · 1/10	1.3 ± 0.1	h ≈ 0.00972
Truncated root	ν(5/3)	1.290994448736	[12/10, 14/10]
Dial denominators	$2^5\cdot22$, $3^5\cdot22$, $7^5\cdot22$	704, 5346, 369754	proved

Appendix B — Manifests, hashes, seeds

Every harness run writes a `MANIFEST.toml` recording the effective configuration with provenance (which entry point set which field), SHA-256 hashes of the produced artifacts, and the seed derivation; `data/build_manifest.json` pins the audit counts (24 modules, 145 theorems, 5108 lines, 33 + 62 headline declarations, build steps 2894/2885, exit 0). The master seed 1993 derives per-task seeds as $(1993 + \texttt{0x9E37}\cdot t) \bmod 2^{32}$, so any report in `results/run_*/` can be replayed to the byte. The reproduction commands, in the order a skeptical reader should run them:

```
cd lean && lake exe cache get! && lake build          # 2 894 steps, exit 0
lake env lean AxiomAudit.lean                         # 33 declarations
lake env lean Satellite/AxiomAudit.lean               # 62 declarations
python scripts/compute_all.py                         # → data/
python scripts/certify.py emit && python scripts/certify.py check
python scripts/septinal_verify.py --ci --preset quick --task all
python scripts/septinal_verify.py --self-test --props --contract
julia   scripts/septinal_verify.jl  --self-test
python scripts/verify.py                              # VERIFY: ALL GREEN
```

Appendix C — The Δ-list: from the initial wing to this edition

The package keeps its history honestly. The table records every structural change between the initial upload of the wing and the present edition — the philosophical contract of the house is that deltas are documented, not absorbed silently. The `data/` artifacts are unchanged by every row except the first: no constant moved.

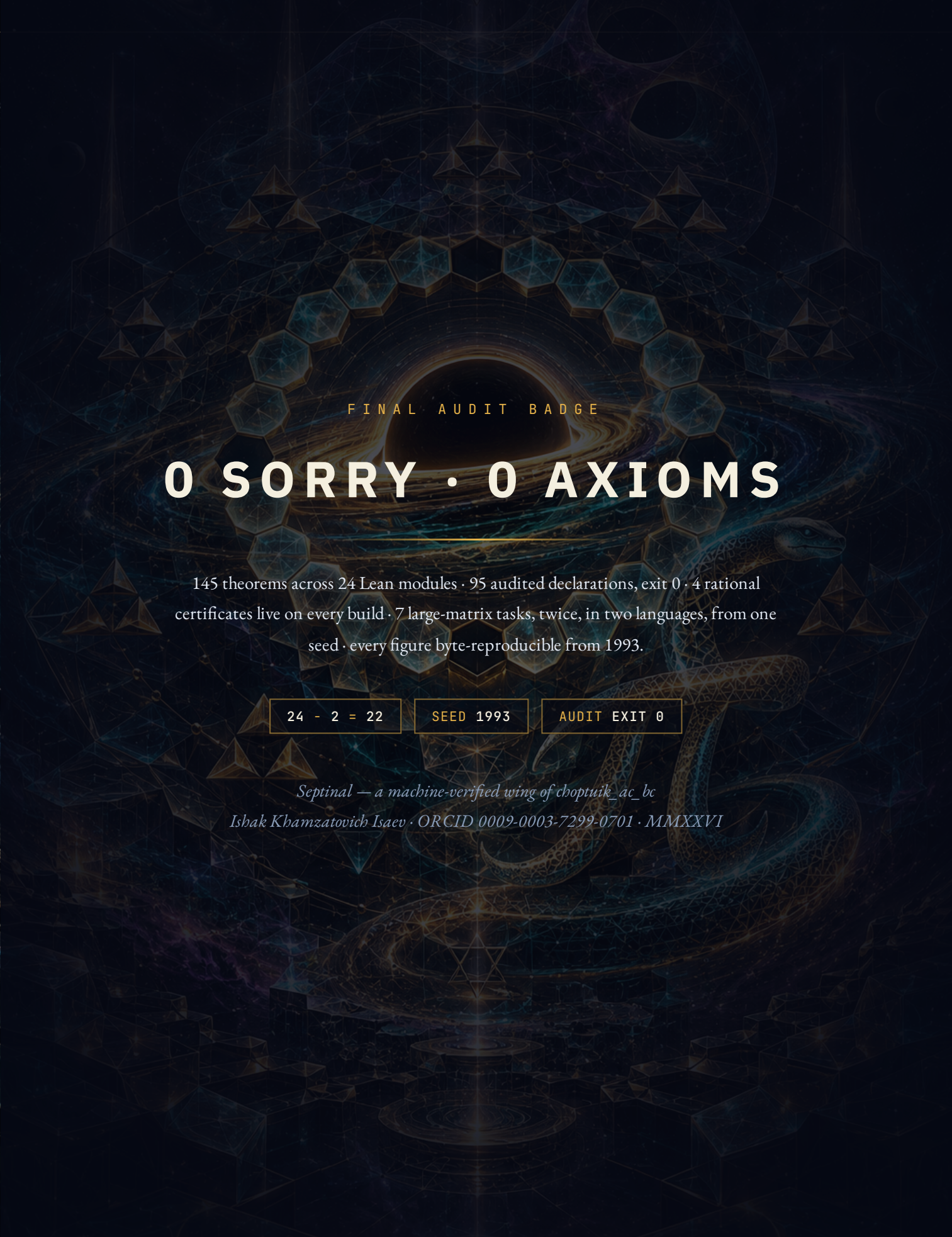
Table C.1 — structural deltas, v0.9 → v1.0

Δ	V0.9 (INITIAL WING)	V1.0 (THIS EDITION)	WHERE
1	10 project axioms	0 — each axiom discharged into proofs; only the mathematical trio remains	`AxiomAudit.lean` × 2 · `AUDIT_EXIT=0`
2	universality stated in raw form	repaired statement in the Scaling module	`Choptuik/Scaling.lean`
3	task plots rendered before series were added — committed PNGs carried empty axes	rendering deferred to report finish; every series reaches the PNG/SVG	`septinal_verify.py` · `.jl`
4	bitmap-glyph tracking 1.7 — legend text collided	tracking 2.8; legend card padded; committed plots legible	`SEPTPlot`, both twins
5	—	this monograph: 8 parts, 7 part plates, the cabinet, the claim index	`monograph/SEPTINAL_Monograph.pdf`

Rows 3 and 4 were found while typesetting this document — the monograph is itself a verification tool: setting the harness's own outputs into print is what exposed the defects. This is the house style: *machine-readable or it did not happen*, applied to the machine's own figures.

Colophon

This monograph was typeset as a single flowing HTML document and rendered to vector PDF at a 200 × 260 mm trim size; the cover and closing plates use the package's cosmic-baroque key artwork, and the mathematical formulas are rendered from the same LaTeX source notation the Lean docstrings use. Body text is set in EB Garamond, headings in IBM Plex Sans, and code in JetBrains Mono; the palette is the package's own — midnight `#070A14`, gold `#E8B44A`, teal glass `#4CC9F0`, emerald audit `#57CC99`, violet nebula `#9B8CDE`. The eight Part figures are the committed monograph renditions from `figures/monograph_en/`; the twenty-one task plots are the verbatim outputs of one committed `--ci --preset quick --task all` run of the Python twin, both twins finishing 7/7 PASS. Every number in every table traces to `data/`, and the whole artifact regenerates from seed 1993. The package is a machine-verified wing of `choptuik_ac_bc`; correspondence: GitHub `@wild8highlander`, ORCID 0009-0003-7299-0701.


FINAL AUDIT BADGE
0 SORRY · 0 AXIOMS
145 theorems across 24 Lean modules · 95 audited declarations, exit 0 · 4 rational certificates live on every build · 7 large-matrix tasks, twice, in two languages, from one seed · every figure byte-reproducible from 1993.
24 - 2 = 22
SEED 1993
AUDIT EXIT 0
Septinal — a machine-verified wing of choptuik_ac_bc
Ishak Khamzatovich Isaev · ORCID 0009-0003-7299-0701 · MMXXVI